

Better Bounds for Finding Fixed-Degree Isogenies via Coppersmith’s Method

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Abstract. The hardness of finding isogenies of degree d between supersingular elliptic curves is a fundamental assumption in isogeny-based cryptography. Let E_1 and E_2 be supersingular elliptic curves defined over $\mathbb{F}_{p^2}$, and let d be a smooth integer. At CRYPTO 2024, Benčina et al. proposed an algorithm with time complexity $\tilde{O}(\max\{p^{1/2}, d/p^{5/8}\})$ in the classical setting and $\tilde{O}(\max\{p^{1/4}, d^{1/2}/p^{1/4}\})$ in the quantum setting. In this work, we first observe that their analysis omits a sub-exponential factor $\exp(O(\log^{3/4} p))$. We then improve their result to $\tilde{O}(\max\{p^{1/2}, \exp(O(\log^{4/5} p)) \cdot d/p^{2/3}\})$ classically and $\tilde{O}(\max\{p^{1/4}, \exp(O(\log^{4/5} p)) \cdot d^{1/2}/p^{1/3}\})$ quantumly. Our approach relies on small-root bounds for Coppersmith’s method applied to a four-variable integer equation. To this end, we adapt the explicit asymptotic formulas for small-root bounds introduced by Feng et al. (CRYPTO 2025) in the modular setting to the integer setting. As an additional application, we strengthen the attack of Benčina et al. on the SIDH signature scheme by Basso et al. (ACNS 2024). We expect that these refined techniques for Coppersmith’s method will be valuable for further post-quantum cryptanalysis.

Keywords: Coppersmith’s method, Isogeny computation, Post-quantum cryptography

1 Introduction

In recent years, isogeny-based cryptography has emerged as a promising post-quantum candidate. The fundamental problem in isogeny-based cryptography is the isogeny problem. Given two supersingular elliptic curves E_1 and E_2 defined

over $\mathbb{F}_{p^2}$, the problem is to find an isogeny $\varphi : E_1 \rightarrow E_2$. For several isogeny-based schemes, attacking them requires finding an isogeny whose degree has a prescribed form or lies within a specific bound [1, 8, 24, 42]. It is therefore natural to study the problem of finding an isogeny between E_1 and E_2 of a specific degree d , given the promise that it exists. We call this the *fixed-degree isogeny problem* (*d-Isogeny*).

Without the restriction on the degree, the isogeny problem is known to be polynomial-time equivalent to the endomorphism ring problem (**EndRing**). However, no general reduction is known from the fixed-degree variant to the endomorphism ring problem. Even if we are given $\text{End}(E_1)$ and $\text{End}(E_2)$, there is no general polynomial time algorithm to compute a fixed-degree isogeny. We only have a poly-time reduction in certain regimes:

- When $d < \sqrt{p}$, the fixed-degree isogeny is typically the shortest isogeny between E and E' , and the problem can be solved using lattice reduction [25]. The approach can be generalized to slightly larger d by exhaustive search.
- Heuristically, when $d > p^{9/2}$ and d is smooth, the problem can be solved using the KLPT algorithm [26, 34]. When the endomorphism ring of one of the curves is a special extremal order, the bound can be improved to $d > p^3$.

In the intermediate range $\sqrt{p} < d < p^3$, the relationship with the endomorphism ring problem remains unclear. This prompted Basso et al. [7] to ask: “Can we construct a secure cryptographic protocol where the endomorphism rings of all curves are public?” Equivalently, we can ask: Given the endomorphism ring of two curves, how hard is it to compute a fixed-degree isogeny between them?

At CRYPTO 2024, Benčina et al. [10] investigated the hardness of the fixed-degree isogeny problem for d in the intermediate range. They improved upon the state of the art, both in the classical and quantum settings, under certain average-case assumptions. In particular, they showed that for $d < p$, the classical asymptotic complexity of finding a d -isogeny between two random supersingular curves is the same as computing their endomorphism rings, with an analogous result holding for quantum algorithms.

The approach of Benčina et al. begins by computing the endomorphism rings of the given curves. Once these rings are known, the fixed-degree isogeny problem can be transferred to the quaternion world via the Deuring correspondence [19]. In the quaternion world, the problem reduces to solving a norm equation in a four-dimensional lattice. Concretely, this Diophantine equation has the form

$$\mathbf{x}^t \mathbf{G} \mathbf{x} = d,$$

where $\mathbf{G} \in \mathbb{Z}^{4 \times 4}$ is a Gram matrix for the lattice and $\mathbf{x} = (x_1, x_2, x_3, x_4)^t \in \mathbb{Z}^4$. They then use Coppersmith’s method to solve this equation.

Interestingly, while Coppersmith’s method is typically used to find small roots, in this setting one can recover all roots if d satisfies certain conditions. By applying LLL to the Gram matrix $\mathbf{G}$ and obtaining a unimodular matrix $\mathbf{U}$, they show that for the root $(x_1^{(0)}, \dots, x_4^{(0)})$, each component of $(x_1^{(0)}, \dots, x_4^{(0)}) \mathbf{U}^{-t}$ is bounded in size. Hence, one can apply Coppersmith’s method to recover $(x_1^{(0)}, \dots, x_4^{(0)}) \mathbf{U}^{-t}$ and subsequently compute $(x_1^{(0)}, \dots, x_4^{(0)})$.

Contributions. The approach of Benčina et al. involves guessing one or two variables to reduce the problem to a two- or three-variable instance of Coppersmith’s method [9, 15, 16]. While it is natural to attempt applying Coppersmith’s method directly to the four-variable equation in order to obtain tighter bounds, deriving explicit asymptotic bounds in this setting is non-trivial, as it requires more involved combinatorial computations.

Recently, Feng et al. [22] introduced tools from additive combinatorics into the analysis of asymptotic bounds for Coppersmith’s method, thereby motivating the extension of these techniques to the integer setting. However, this extension is not straightforward. For a k -variable polynomial, as shown in [22], the dimension and determinant of the lattice are polynomials in m of degree k and $k + 1$, respectively. In contrast, in the integer setting, both the dimension and the determinant turn out to be degree- k polynomials.

In this work, we continue this line of research with the following contributions:

Explicit formulas for Coppersmith’s method over integer polynomial equations. Motivated by [22], we derive explicit formulas for the asymptotic bounds of Coppersmith’s method applied to integer polynomial equations, focusing on the Jochemsz–May strategy [31]. The key distinction lies in the computation of the lattice determinant.

In [22], the determinant consists of two types of contributions: powers of the modulus and powers of the upper bounds X_j for $j = 1, \dots, k$. The latter are given by the sum of the degrees in x_j of all monomials in f^m for some integer m , which yields a polynomial $p_j(m)$ of degree $k + 1$ asymptotically. For asymptotic bounds, only the leading coefficient of p_j is relevant.

In the integer setting, however, the lattice is constructed differently: now the power of X_j in the determinant is given asymptotically by $p_j(m) - p_j(m - 1)$. Thus, the analysis again reduces to studying the leading coefficient. In particular, the leading coefficient of $p_j(m) - p_j(m - 1)$ is $(k + 1)$ times that of $p_j(m)$.

Furthermore, to address scenarios involving partial information leakage, we generalize the Jochemsz–May strategy from a single polynomial equation to systems of polynomial equations. However, in our strategy, we impose restrictions on the input polynomials to ensure that Coppersmith’s method produces $k - 1$ polynomials that are algebraically independent of the input polynomials. We regard the removal of these restrictions as an important open problem for future research.

Solving quadratic forms for Gram matrices. As an application, we analyze quadratic forms of the type $f_{\mathbf{G}} = \mathbf{x}^t \mathbf{G} \mathbf{x} - d \in \mathbb{Z}[x_1, \dots, x_k]$, where $\mathbf{G} = \mathbf{B}^t \mathbf{B}$ for some $\mathbf{B} \in \mathbb{Z}^{k \times k}$. For simplicity, we assume that the LLL-reduced basis of $\mathbf{B}$ has uniform Euclidean norm, i.e., approximately $\det(\mathbf{B})^{1/k}$. Roughly, we show that if $\det(\mathbf{G})^{1/k} < d < \det(\mathbf{G})^{1/(k-1)-\varepsilon}$ for some constant $\varepsilon > 0$, then all roots of $f_{\mathbf{G}}$ can be found in polynomial time.⁵

⁵ See Theorem 4 for the formal statement.

Improved bounds for finding smooth fixed-degree isogenies. We apply our explicit formulas to the polynomial equations related to d -Isogeny problem. Under the same average-case assumptions as in [10], we show that d -Isogeny reduces to EndRing when $d < p^{2/3-\varepsilon}$ for some arbitrarily small constant $\varepsilon > 0$. In other words, given the endomorphism ring of the domain E_1 and codomain E_2 , we can compute a degree d isogeny $\varphi : E_1 \rightarrow E_2$ in polynomial time, if it exists. This reduction is made possible by the fact that four-variable Coppersmith does not require any guessing. To apply bivariate or trivariate Coppersmith, one or two variables must be guessed, which we do not know how to do in polynomial time.

Additionally, we improve on the best known algorithm for solving the d -Isogeny problem when d is smooth. Using trivariate Coppersmith, Benčina et al. [10] are able to classically solve the d -Isogeny problem for smooth d in time

$$\tilde{O}(\max\{p^{1/2}, d/p^{5/8-\varepsilon}\},$$

and quantumly in time

$$\tilde{O}(\max\{p^{1/4}, d^{1/2}/p^{1/4-\varepsilon}\}).$$

Using our four-variable Coppersmith variant, we improve the classical runtime to

$$\tilde{O}(\max\{p^{1/2}, d/p^{2/3-\varepsilon}\}),$$

and the quantum runtime to

$$\tilde{O}(\max\{p^{1/4}, d^{1/2}/p^{1/3-\varepsilon}\}).$$

Finally, we address some minor issues in [10] in Appendix D. First, their analysis omits the ε term. For multivariate Coppersmith, the ε -term is inherent and cannot be easily eliminated [40]. In Appendix D.1, we show that setting $\varepsilon = 0$ requires increasing the asymptotic runtime by a sub-exponential factor $\exp(O(\log^{4/5} p))$. Second, they employ the trivariate Coppersmith method without explicitly stating the required algebraic independence assumption (our Assumption 1). Third, when applying bivariate or trivariate Coppersmith, the input polynomial must be irreducible. Although the initial polynomial in four variables is irreducible, one must verify that after guessing one or two variables, the resulting polynomial is still irreducible. While this should hold most of the time, it might fail in exceptional cases. We give two concrete examples in Appendix D.2. With four variable Coppersmith, we avoid this issue.

Related work. Recently, Eriksen and Leroux [21] proposed new algorithms for embedding quadratic orders into maximal orders and applied them to finding connecting ideals of fixed norm, which in turn can be used to compute fixed-degree isogenies. They obtained two main results: (1) an $O(p^{2/3})$ algorithm for finding a connecting ideal of norm d , provided one exists, and (2) a polynomial-time algorithm for finding a connecting ideal of norm $d < p^{2/3}$, assuming that one of the maximal orders contains a small-norm non-scalar element and that integer factorization is available. Consequently, they can solve the fixed-degree

isogeny problem in polynomial time for $d < p^{2/3}$, provided the endomorphism ring is known.

In Theorem 5 of our work, we show that the fixed-degree isogeny problem can be solved in polynomial time for $d < p^{2/3-\varepsilon}$ for any constant $\varepsilon > 0$, again assuming the endomorphism ring is known. The difference between our result and that of [21] lies in the ε -term in our bound, which could be removed at the cost of an additional $\exp(O(\log^{4/5} p))$ factor. Their result, in contrast, relies on the availability of a factorization oracle and the existence of an element of unusually small norm in one of the endomorphism rings. Moreover, the underlying assumptions differ: we rely on Assumption 1, which is widely used in Coppersmith-type analyses, whereas they (see Proposition 9 in [21]) employ a heuristic assumption that integers do not have too many prime factors.

Organization. The rest of the paper is organized as follows. Section 2 covers background on Linear Algebra, Coppersmith’s method, isogenies and combinatorics. Section 3 derives explicit bounds for Coppersmith’s over integer polynomials. Section 4 analyzes quadratic forms with Gram matrices. Section 5 applies Coppersmith’s method to the fixed-degree isogeny problem, obtaining our main results. Section 6 presents the performance of our algorithm under different parameter settings. Section 7 concludes the paper.

2 Preliminaries

We use $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$ to denote the ring of integers, the field of rationals, and the field of real numbers, respectively. We also use $\mathbb{Z}_{>0}$ to denote the set of positive integers. Vectors are represented by lowercase bold letters (e.g., $\mathbf{v}$) and are defined as column vectors throughout this paper. Matrices are denoted by uppercase bold letters (e.g., $\mathbf{A}$). For complexity notation, we use $O(\cdot)$ for asymptotic upper bounds and $\Theta(\cdot)$ for asymptotic tight bounds, and $\text{polylog}(\cdot)$ for polylogarithmic factors. When omitting factors polynomial in $\log p$, where p is the characteristic of the finite field under consideration, we abbreviate $O(x \cdot \text{polylog } p)$ by $\tilde{O}(x)$.

We call a positive integer *smooth* if all its prime factors are bounded by a polynomial in $\log p$. We use $\|\cdot\|_2$ for the Euclidean norm, $\|\cdot\|_\infty$ for the infinity norm, and $|\cdot|$ for the cardinality of a set (e.g., $|S|$ denotes the size of a set S).

We introduce Coppersmith’s method and basic lattice theory in Section 2.2, provide background on isogenies in Section 2.4, Section 2.5, and Section 2.6, and review useful results from additive combinatorics in Section 2.3.

2.1 Polynomials, Lattices

Let $f(x_1, \dots, x_k) = \sum a_{i_1 \dots i_k} x_1^{i_1} \dots x_k^{i_k} \in \mathbb{Z}[x_1, \dots, x_k]$. We begin by introducing some notation. We call $x_1^{i_1} \dots x_k^{i_k}$ a monomial of f if $a_{i_1 \dots i_k} \neq 0$. The total degree of f is defined as

$$\max_{a_{i_1, \dots, i_k} \neq 0} \{i_1 + \dots + i_k\},$$

which we write as $\deg(f)$.

The set of all monomials of f is called the support of f , which is defined as $\text{supp}(f)$. We also consider a geometric version of $\text{supp}(f)$, which is defined as follows:

$$A(f) = \{(i_1, \dots, i_k) | x_1^{i_1} \dots x_k^{i_k} \in \text{supp}(f)\} \subseteq \mathbb{Z}^k.$$

We use $N(\cdot)$ for the convex hull. The convex hull of $A(f)$ is called the Newton polytope of f , which we write as $N(f)$ for simplicity.

In this paper, we also consider the infinity norm of f , that is $\|f\|_\infty$, which is defined as

$$\|f\|_\infty = \max_{x_1^{i_1}, \dots, x_k^{i_k} \in \text{supp}(f)} \{|a_{i_1 \dots i_k}|\}.$$

An irreducible polynomial is, roughly speaking, a polynomial that cannot be factored into the product of two non-constant polynomials. In this paper, when we refer to a polynomial as irreducible over the integers, we additionally require that its coefficients are relatively prime as a set (i.e., their greatest common divisor is 1).

Lemma 1. *Let $f(x_1, \dots, x_k) \in \mathbb{Z}[x_1, \dots, x_k]$ be irreducible over the integers. For any integer $a \in \mathbb{Z}$, the polynomial*

$$f(x_1 + a, x_2, \dots, x_k)$$

is also irreducible in $\mathbb{Z}[x_1, \dots, x_k]$.

Proof. Consider the ring automorphism

$$\Phi_a : \mathbb{Z}[x_1, \dots, x_k] \rightarrow \mathbb{Z}[x_1, \dots, x_k], \quad g(x_1, \dots, x_k) \mapsto g(x_1 + a, x_2, \dots, x_k),$$

whose inverse is Φ_{-a} . Suppose for contradiction that

$$\Phi_a(f) = g_1 g_2$$

with $g_1, g_2 \notin \{\pm 1\}$. Applying Φ_{-a} gives a nontrivial factorization

$$f = \Phi_{-a}(g_1) \Phi_{-a}(g_2),$$

contradicting the irreducibility of f . Hence $f(x_1 + a, x_2, \dots, x_k)$ is irreducible. $\square$

This lemma is useful when enumerating the most significant bits in Copper-smith's method.

Now we give some definitions and facts about lattices.

Let $K \subseteq \mathbb{R}$ be a field. A quadratic space (V, Q) is a K -vector space V of finite dimension m together with a quadratic form $Q : V \rightarrow K$. The quadratic form Q has the associated bilinear form

$$\langle \mathbf{x}, \mathbf{y} \rangle = \frac{1}{2} (Q(\mathbf{x} + \mathbf{y}) - Q(\mathbf{x}) - Q(\mathbf{y})).$$

We say that Q is positive definite if $Q(\mathbf{x}) > 0$ whenever $\mathbf{x} \neq \mathbf{0}$. If Q is positive definite, then the associated bilinear form is an inner product.

A lattice in (V, Q) is a subgroup $\mathcal{L} \subseteq V$ of rank $n \leq m$. A $\mathbb{Z}$ -basis of $\mathcal{L}$ is a collection of linearly independent vectors $\mathbf{b}_1, \dots, \mathbf{b}_n \in V$ such that $\mathcal{L} = \mathbb{Z}\mathbf{b}_1 + \dots + \mathbb{Z}\mathbf{b}_n$. We let $\dim(\mathcal{L}) = m$ and $\text{rank}(\mathcal{L}) = n$. In the rest of this document, we will only work with lattices of full rank, with $n = m$.

Given a $\mathbb{Z}$ -basis $\mathbf{b}_1, \dots, \mathbf{b}_n$ of $\mathcal{L}$, its Gram matrix is

$$\mathbf{G} = (\langle \mathbf{b}_i, \mathbf{b}_j \rangle)_{1 \leq i, j \leq n} = (\mathbf{b}_i^t \cdot \mathbf{b}_j)_{1 \leq i, j \leq n}.$$

It is positive definite if and only if f is positive definite. The determinant of $\mathcal{L}$ is defined as $\det(\mathcal{L}) = \det(\mathbf{G})$, and is independent of the basis chosen. The discriminant of $\mathcal{L}$ is defined as $\text{disc}(\mathcal{L}) = 2^n \det(\mathcal{L})$.

When $V = \mathbb{R}^n$ and Q is the standard Euclidean norm, we say that $\mathcal{L}$ is an Euclidean lattice. For an Euclidean lattice with basis $\mathbf{b}_1, \dots, \mathbf{b}_n$, the Gram matrix $\mathbf{G} = \mathbf{B}^t \mathbf{B}$, where the columns of the matrix $\mathbf{B}$ are the basis vectors.

In lattice theory, the Shortest Vector Problem (SVP) consists of finding the non-zero vector with minimal Euclidean norm, and it is proven to be NP-hard under randomized reductions [2]. However, finding a relatively short vector is not as difficult and can be performed in polynomial time using the classical LLL algorithm [35].

Lemma 2 (LLL Basis Reduction). *Let $\mathcal{L} \subseteq \mathbb{Z}^n$ be a lattice. The LLL algorithm outputs a basis $(\mathbf{b}_1, \dots, \mathbf{b}_n)$ of $\mathcal{L}$ satisfying*

$$\|\mathbf{b}_i\|_2 \leq 2^{\frac{n(n-1)}{4(n+1-i)}} \det(\mathcal{L})^{\frac{1}{n+1-i}},$$

for all $i = 1, \dots, n$, in polynomial time.

The Gram-Schmidt orthogonalization (GSO) of a basis $\mathbf{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ is defined recursively by

$$\mathbf{b}_i^* = \mathbf{b}_i - \sum_{j < i} \mu_{i,j} \mathbf{b}_j^*, \quad \text{where} \quad \mu_{i,j} = \frac{\langle \mathbf{b}_i, \mathbf{b}_j^* \rangle}{\langle \mathbf{b}_j^*, \mathbf{b}_j^* \rangle}.$$

A basis $\mathbf{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ is δ -LLL reduced if (i) $|\mu_{i,j}| \leq 1/2$ for all $j < i$, and (ii) $\delta \|\mathbf{b}_i^*\|_2^2 \leq \|\mu_{i+1,i} \mathbf{b}_i^* + \mathbf{b}_{i+1}^*\|_2^2$ for all i . In this paper we choose $\delta = 3/4$. Throughout this paper we set $\delta = 3/4$.

A Minkowski-reduced basis $\mathbf{b}_1, \dots, \mathbf{b}_n$ of a lattice $\mathcal{L}$ is a basis of $\mathcal{L}$ that is as close to orthogonal as possible, with short, well-separated vectors. We follow the convention $\|\mathbf{b}_1\|_2 \leq \|\mathbf{b}_2\|_2 \leq \dots \leq \|\mathbf{b}_n\|_2$. The successive minima $\lambda_1, \dots, \lambda_n$ of $\mathcal{L}$ are defined as the minimal radius such that each λ_i is the smallest radius containing i linearly independent lattice vectors.

2.2 Coppersmith's method

In 1996, Coppersmith introduced a method for finding small roots of univariate modular polynomial equations [14, 15] or bivariate integer polynomial equations [13, 15] using the LLL algorithm. Since then, the method has found many different applications in public-key cryptography [23, 36, 37, 40, 46–48].

Given irreducible $f(x_1, \dots, x_k) \in \mathbb{Z}[x_1, \dots, x_k]$ and some bounds $X_1, \dots, X_k \in \mathbb{Z}_{>0}$, the goal of Coppersmith's method is to find all roots $(x_1^{(0)}, \dots, x_k^{(0)})$ of f over $\mathbb{Z}$ such that $|x_i^{(0)}| < X_i$ for $i = 1, \dots, k$.

The intuition behind the method is to first choose a modulus R and then construct a family of polynomials $G = \{g_i(x_1, \dots, x_k)\}$ that share the same root with $f \bmod R$. Then use the coefficient vector of $g_i(x_1 X_1, \dots, x_k X_k)$ as the basis to construct the lattice $\mathcal{L}$. Using the LLL algorithm (Lemma 2) we can find some relatively short vector in $\mathcal{L}$. The following Lemma 3 show that if we can find $k - 1$ short enough vectors in $\mathcal{L}$, we can have $k - 1$ polynomials $h_1, \dots, h_{k-1}$ which share the same root with f over $\mathbb{Z}$. Then we can compute the target roots.

Lemma 3 (Howgrave-Graham [29]). *Let $h(x_1, \dots, x_k) \in \mathbb{Z}[x_1, \dots, x_k]$ be a polynomial with at most ω monomials, and let R be a positive integer. If there exist integers $(u_1, \dots, u_k)$ satisfying the following two conditions:*

1. $h(u_1, \dots, u_k) \equiv 0 \pmod R$,
2. *there exist positive integers $X_1, \dots, X_k$ such that $|u_i| < X_i$ for $i = 1, \dots, k$, and*

$$\|h(x_1 X_1, \dots, x_k X_k)\|_2 < \frac{R}{\sqrt{\omega}},$$

then $h(u_1, \dots, u_k) = 0$ holds over $\mathbb{Z}$.

To make the reduced basis of LLL in Lemma 2 satisfy the condition in Lemma 3, we require the lattice $\mathcal{L}$ to satisfy the following inequality.

$$2^{\frac{\dim(\mathcal{L})(\dim(\mathcal{L})-1)}{4(\dim(\mathcal{L})+1-i)}} \det(\mathcal{L})^{\frac{1}{\dim(\mathcal{L})+1-i}} \leq \frac{R}{\sqrt{\dim(\mathcal{L})}}, \quad (1)$$

for $i = 1, \dots, k - 1$.

Remark 1. *The choice of R ensures that the h_i are independent of f . Details can be found in [31].*

However, we cannot ensure that the h_i are pairwise algebraically dependent. So we need the following (well-established) heuristic for $k \geq 3$, which holds for most instances arising in practice [22, 28, 36, 38, 39, 46, 47].

Assumption 1. *The polynomials obtained from the LLL-reduced basis in Coppersmith's method generate an ideal corresponding to a zero-dimensional variety.*

2.3 Additive Combinatorics

We now introduce several useful results from additive combinatorics that can be applied to compute asymptotic bounds for Coppersmith's method [22, 46]. We mainly consider finite subsets $A \subseteq \mathbb{Z}^k$ that contain $\mathbf{0} = (0, \dots, 0) \in \mathbb{Z}^k$.

Definition 1 (Index of A). Let $\mathbb{Z}^k(A)$ denote the subgroup generated by

$$\left\{ \sum n_i \mathbf{a}_i \mid \mathbf{a}_i \in A, n_i \in \mathbb{Z}, \sum n_i = 0 \right\}.$$

If $\mathbb{Z}^k(A)$ has finite index $\text{ind } A$ in $\mathbb{Z}^k$, then $\text{ind } A$ is called the *index* of A .

Definition 2 (Minkowski Sum). For sets $A, B \subset \mathbb{Z}^k$, their *Minkowski sum* is defined as

$$A + B = \{\mathbf{a} + \mathbf{b} \mid \mathbf{a} \in A, \mathbf{b} \in B\}.$$

For $m \in \mathbb{N}$, denote by mA the sum of m copies of A . We are interested in the cardinality of mA .

Lemma 4 (Corollary 2, [32]). *Suppose $\mathbf{0} \in A \subseteq \mathbb{Z}^k$ and A has finite index $\text{ind } A$ in $\mathbb{Z}^k$. Then there exists an integer M such that for all $m > M$, the number of integer points in mA is a polynomial in m of degree k . Specifically, the leading coefficient equals the integral of 1 over $N(A)$ divided by $\text{ind } A$:*

$$\lim_{m \rightarrow \infty} \frac{|mA|}{m^k} = \frac{\int_{N(A)} 1 dV}{\text{ind } A}.$$

We observe that

$$|mA| = \sum_{(i_1, \dots, i_k) \in mA} 1.$$

Thus, $|mA|$ can be viewed as the sum of the constant function 1. We also consider sums of homogeneous polynomials ϕ , such as the projection polynomial ϕ_{j_0} :

$$\phi_{j_0} : \mathbb{Z}^k \rightarrow \mathbb{Z}, (i_1, \dots, i_k) \mapsto i_{j_0}.$$

The sum of homogeneous polynomials has been analyzed in [12], and further developed in [3–6]. We use the restatement of Lemma 10 from [22]:

Lemma 5 (Lemma 10, [22]). *Suppose $\mathbf{0} \in A \subseteq \mathbb{Z}^k$ and A has finite index $\text{ind } A$ in $\mathbb{Z}^k$. Then there exists M such that for all $m > M$, the sum*

$$\sum_{(i_1, \dots, i_k) \in mN(A) \cap \mathbb{Z}^k} i_{j_0}$$

is a polynomial in m of degree $k + 1$. Specifically, the leading coefficient equals the integral of x_{j_0} over $N(A)$ divided by $\text{ind } A$:

$$\lim_{m \rightarrow \infty} \frac{\sum_{(i_1, \dots, i_k) \in mA} i_{j_0}}{m^{k+1}} = \frac{\int_{N(A)} x_{j_0} dV}{\text{ind } A}.$$

Beyond sums over mA , we also study sums over $mN(A) \cap \mathbb{Z}^k$. By Ehrhart's theorem [20], this quantity is polynomial in m for any integral polytope $N(A) \subset \mathbb{R}^k$. We restate Theorem 1 from [18] as Lemma 6:

Lemma 6 (Theorem 1, [18]). *Suppose $\mathbf{0} \in A \subseteq \mathbb{Z}^k$, and let N be the convex hull of A . Then the number of integer points in $mN(A)$ is polynomial in m of degree k . Specifically, the leading coefficient equals the integral of 1 over $N(A)$:*

$$\left| \sum_{(i_1, \dots, i_k) \in mN(A) \cap \mathbb{Z}^k} 1 \right| = \int_{N(A)} 1 dV m^k + o(m^k).$$

We also consider sums involving ϕ_{j_0} :

Lemma 7 (Proposition 4.1, [12]). *Suppose $\mathbf{0} \in A \subseteq \mathbb{Z}^k$, and let N be the convex hull of A . Then*

$$\sum_{(i_1, \dots, i_k) \in mN(A) \cap \mathbb{Z}^k} i_{j_0}$$

is polynomial in m of degree $k+1$. Specifically, the leading coefficient equals the integral of x_{j_0} over $N(A)$:

$$\left| \sum_{(i_1, \dots, i_k) \in mN(A) \cap \mathbb{Z}^k} i_{j_0} \right| = \int_{N(A)} x_{j_0} dV m^{k+1} + o(m^{k+1}).$$

2.4 Isogenies

Let E, E' be elliptic curves defined over a finite field $\mathbb{F}_q$. An isogeny $\varphi : E \rightarrow E'$ is a non-constant rational map that also defines a group homomorphism $E(\overline{\mathbb{F}_q}) \rightarrow E'(\overline{\mathbb{F}_q})$. The degree $\deg(\varphi)$ is its degree as a rational map, and thus a non-negative integer. It is multiplicative under composition: for an isogeny $\psi : E' \rightarrow E''$, $\deg(\psi \circ \varphi) = \deg(\psi) \cdot \deg(\varphi)$. An isogeny of degree 1 is an isomorphism. The isomorphism class of E is characterized by its j -invariant $j(E) \in \mathbb{F}_q$. We denote the set of isogenies from E to E' by $\text{Hom}(E, E')$.

An isogeny from a curve E to itself is called an endomorphism of E . Together with the zero map, the endomorphisms of E form a ring $\text{End}(E)$. For any integer n , the scalar multiplication map $[n] : E \rightarrow E, P \mapsto nP$ is an endomorphism of E of degree n^2 . For any isogeny $\varphi : E \rightarrow E'$, its dual is the unique isogeny $\hat{\varphi} : E' \rightarrow E$ such that $\varphi \circ \hat{\varphi} = [\deg \varphi]$ and $\hat{\varphi} \circ \varphi = [\deg \varphi]$. The trace of an endomorphism $\alpha \in \text{End}(E)$ is $\alpha + \hat{\alpha} \in \mathbb{Z}$.

We say that E is *supersingular* if $\text{End}(E)$ is a $\mathbb{Z}$ -lattice of rank 4. A supersingular curve E defined over characteristic p is always isomorphic to a curve defined over $\mathbb{F}_{p^2}$.

2.5 Quaternion algebras

For any $a, b \in \mathbb{Q}^\times$, the quaternion algebra $B = \left(\frac{a, b}{\mathbb{Q}} \right)$ is the ring generated by the $\mathbb{Q}$ -basis $1, i, j, ij$ satisfying

$$i^2 = a, j^2 = b, ij = -ji.$$

For a prime p , we say that B is ramified at p if $B \otimes \mathbb{Q}_p \not\cong M_2(\mathbb{Q}_p)$. It is ramified at ∞ if $B \otimes \mathbb{R} \not\cong M_2(\mathbb{R})$. For a prime $p > 2$, we define $B_{p,\infty}$ as a quaternion algebra ramified at p and ∞ (and unramified at all other primes). It is unique up to isomorphism.

$$B_{p,\infty} \cong \left(\frac{-q, -p}{\mathbb{Q}} \right), \text{ where } q = \begin{cases} 1 & \text{if } p \equiv 3 \pmod{4} \\ 2 & \text{if } p \equiv 5 \pmod{8} \\ q_p & \text{if } p \equiv 1 \pmod{8} \end{cases}$$

and q_p is any prime that is 3 mod 4 and is not a square modulo p .

Any quaternion algebra B has a standard involution

$$\alpha = t + xi + yj + zij \mapsto \bar{\alpha} = t - xi - yj - zij.$$

It induces the reduced trace $\text{trd}(\alpha) = \alpha + \bar{\alpha} = 2t$ and the reduced norm $\text{nrd}(\alpha) = \alpha \bar{\alpha} = t^2 - ax^2 - by^2 + abz^2$. The reduced norm is a quadratic form on B , with associated bilinear form

$$\langle \alpha, \beta \rangle = \frac{1}{2} \text{trd}(\alpha \bar{\beta}) = \frac{1}{2} (\text{nrd}(\alpha + \beta) - \text{nrd}(\alpha) - \text{nrd}(\beta)).$$

When B is ramified at ∞ , the reduced norm nrd is positive definite. In this case, the bilinear form is an inner product. For $B_{p,\infty}$, we have the canonical embedding into Euclidean space,

$$\iota : B_{p,\infty} \rightarrow \mathbb{R}^4, 1 \mapsto (1, 0, 0, 0), i \mapsto (0, \sqrt{q}, 0, 0), j \mapsto (0, 0, \sqrt{p}, 0), ij \mapsto (0, 0, 0, \sqrt{pq}).$$

It is an isometry, in the sense that for all $\alpha \in B_{p,\infty}$, $\|\alpha\|^2 = \text{nrd}(\alpha)$, where $\|\cdot\|$ is the Euclidean norm.

An order $\mathcal{O}$ of B is a subring of B that is also a $\mathbb{Z}$ -lattice of rank 4. The elements $\alpha \in \mathcal{O}$ are integral, meaning that $\text{trd}(\alpha) \in \mathbb{Z}$ and $\text{nrd}(\alpha) \in \mathbb{Z}$. An order is maximal if it is not strictly contained in another order. Maximal orders in $B_{p,\infty}$ have $\text{disc}(\mathcal{O}) = p^2$.

A $\mathbb{Z}$ -lattice $\Lambda \subset B$ of rank 4 has an associated left and right order

$$\mathcal{O}_L(\Lambda) = \{\alpha \in B \mid \alpha \Lambda \subseteq \Lambda\} \text{ and } \mathcal{O}_R(\Lambda) = \{\alpha \in B \mid \Lambda \alpha \subseteq \Lambda\}.$$

We say that Λ is an *ideal* if it is contained in its left and right orders. An ideal I is a left $\mathcal{O}_L(I)$ -ideal and a right $\mathcal{O}_R(I)$ -ideal. We say that it is a connecting ideal from $\mathcal{O}_L(I)$ to $\mathcal{O}_R(I)$. If $\mathcal{O}_L(I)$ is a maximal order, then so is $\mathcal{O}_R(I)$. Two ideals I and J are said to be equivalent if there exists a unit $\beta \in B^\times$ such that $I = J\beta$. Equivalent ideals have isomorphic left and right orders.

The reduced norm of an ideal is $\text{nrd}(I) = \gcd \{\text{nrd}(\alpha) \mid \alpha \in I\}$. Its reduced quadratic form is

$$q_I : I \rightarrow \mathbb{Z}, \alpha \mapsto \frac{\text{nrd}(\alpha)}{\text{nrd}(I)}.$$

In $B_{p,\infty}$, an ideal I whose left order is maximal has $\text{disc}(I) = p^2$ with respect to this quadratic form. For a Minkowski-reduced basis $\sigma_1, \dots, \sigma_4$ of I , the Minkowski bound states that (cf. [34])

$$p^2 \leq 16q_I(\sigma_1)q_I(\sigma_2)q_I(\sigma_3)q_I(\sigma_4) \leq 4p^2 \quad (2)$$

2.6 The Deuring correspondence

In [19], Deuring showed that there is a deep connection between supersingular curves over characteristic p and the quaternion algebra $B_{p,\infty}$. For a supersingular curve E , its endomorphism ring $\text{End}(E)$ is isomorphic to a maximal order $\mathcal{O}$ in $B_{p,\infty}$. An endomorphism of degree d and norm t is mapped to an element of $\mathcal{O}$ of reduced norm d and reduced trace t . Conversely, any maximal order $\mathcal{O}$ in $B_{p,\infty}$ corresponds to the endomorphism ring of some supersingular curve.

The Deuring correspondence can be seen as a dictionary between the objects in $B_{p,\infty}$ and the objects in the world of supersingular curves. A supersingular j -invariant (up to Galois conjugacy) corresponds to a maximal order in $B_{p,\infty}$ (up to isomorphism). An isogeny $\varphi : E \rightarrow E'$ of degree d corresponds to an ideal I_φ of reduced norm $\text{nrd}(I) = d$ with $O_L(I) \cong \text{End}(E)$ and $O_R(I) \cong \text{End}(E')$. Two isogenies $\varphi : E \rightarrow E'$ and $\psi : E \rightarrow E'$ correspond to equivalent ideals $I_\varphi \sim I_\psi$. As $\mathbb{Z}$ -lattices, I_φ (with quadratic form q_I) is isomorphic to $\text{Hom}(E, E')$ (with quadratic form $\deg$).

3 Computing Asymptotic Bounds in Coppersmith's Method for Integer Polynomial Equations

We first propose a Newton–polytope version of the Jochemsz–May strategy for integer equations [31], and then present the corresponding theorem for the asymptotic bound on small roots in Coppersmith's method.

3.1 For a Single Polynomial Equation

Let $f(x_1, \dots, x_k) = \sum a_{i_1 \dots i_k} x_1^{i_1} \dots x_k^{i_k} \in \mathbb{Z}[x_1, \dots, x_k]$ be irreducible. We introduce the following notation. Given bounds $X_1, \dots, X_k \in \mathbb{Z}_{>0}$, the goal is to find all integer roots $(x_1^{(0)}, \dots, x_k^{(0)})$ such that $|x_i^{(0)}| < X_i$ for $i = 1, \dots, k$. We denote $W = \|f(x_1 X_1, \dots, x_k X_k)\|_\infty$.

Basic Strategy. We adapt the Jochemsz–May Basic Strategy [31]. Let $m \in \mathbb{N}$. Define

$$\begin{aligned} S &= \{x_1^{i_1} \dots x_k^{i_k} \mid (i_1, \dots, i_k) \in (m-1)A(f)\}, \\ M &= \{x_1^{i_1} \dots x_k^{i_k} \mid (i_1, \dots, i_k) \in mA(f)\}. \end{aligned}$$

Furthermore, set $R = W \prod_{j=1}^k X_j^{l_j}$, where l_j is the largest exponent of x_j among the monomials in S , i.e., $l_j = d_j(m-1)$. To work with a polynomial whose constant term is 1, define $\tilde{f} = a_0^{-1} f \bmod R$, where a_0 is the constant term of

f ⁶. Define the following shift polynomials:

$$\begin{aligned} g &= x_1^{i_1} \cdots x_k^{i_k} \tilde{f}(x_1, \dots, x_k) \prod_{j=1}^k X_j^{l_j - i_j} \quad \text{for } x_1^{i_1} \cdots x_k^{i_k} \in S, \\ \tilde{g} &= x_1^{i_1} \cdots x_k^{i_k} R \quad \text{for } x_1^{i_1} \cdots x_k^{i_k} \in M \setminus S. \end{aligned}$$

All polynomials g and $\tilde{g}$ share the roots of $f \bmod R$. We use the coefficient vectors of $g(x_1 X_1, \dots, x_k X_k)$ and $\tilde{g}(x_1 X_1, \dots, x_k X_k)$ as a basis to construct the lattice $\mathcal{L}$. To obtain $k-1$ polynomials that share the same roots as f over $\mathbb{Z}$ (by Lemma 3), we require $\mathcal{L}$ to satisfy

$$2^{\frac{\dim(\mathcal{L})(\dim(\mathcal{L})-1)}{4(\dim(\mathcal{L})+1-i)}} \det(\mathcal{L})^{\frac{1}{\dim(\mathcal{L})+1-i}} \leq \frac{R}{\sqrt{\dim(\mathcal{L})}}, \quad (3)$$

for $i = 1, \dots, k-1$.

Let $\varepsilon > 0$ be an arbitrarily small error term. We choose the integer m (as a function of $1/\varepsilon$) sufficiently large. For simplicity, we let powers of 2 and factors $\sqrt{\dim(\mathcal{L})}$ that do not depend on R be absorbed into the error term ε , and we use the determinant condition $\det(\mathcal{L}) \leq R^{\dim(\mathcal{L})+2-k}$. A detailed analysis of this ε -term will be provided later.

Write

$$\det(\mathcal{L}) = X_1^{p_1} \cdots X_k^{p_k} R^{p_R},$$

where $p_1, \dots, p_k$ and p_R are functions of m . Then the inequality $\det(\mathcal{L}) \leq R^{\dim(\mathcal{L})+2-k}$ reduces to

$$\prod_{j=1}^k X_j^{p_j} < W^{p_W}, \quad \text{for } p_j = \sum_{x_1^{i_1} \cdots x_k^{i_k} \in M \setminus S} i_j, \quad \text{and } p_W = |S| + 2 - k,$$

from which we derive the success conditions for Coppersmith's method, i.e., bounds on X_j for $j = 1, \dots, k$.

We are typically interested in the behavior as $m \rightarrow \infty$, yielding the following notion.

Definition 3 (Asymptotic Bound). If there exist an integer k_0 and real constants $\sigma_1, \dots, \sigma_k, \sigma_W$ such that

$$\begin{cases} \lim_{m \rightarrow \infty} \frac{p_j}{m^{k_0}} = \sigma_j & \text{for } j = 1, \dots, k, \\ \lim_{m \rightarrow \infty} \frac{p_W}{m^{k_0}} = \sigma_W, \end{cases}$$

then we call $\sigma_1, \dots, \sigma_k, \sigma_W$ the *asymptotic bound*. We use the inequality

$$\prod_{j=1}^k X_j^{\sigma_j} < W^{\sigma_W - \varepsilon}$$

⁶ This requires $a_0 \neq 0$ and $\gcd(a_0, R) = 1$, which can be ensured by a standard transformation; see [16, Appendix A].

as the asymptotic success condition for Coppersmith's method to find all integer roots $(x_1^{(0)}, \dots, x_k^{(0)})$ such that $|x_i^{(0)}| < X_i$ for $i = 1, \dots, k$, where $\varepsilon > 0$ comes from the error term.

We now present explicit formulas for the asymptotic bounds on small roots. For simplicity, we assume f satisfies the property introduced by Feng et al. [22], called *well-defined k -variate*.

Definition 4 (Well-defined k -variate [22]). Let $f(x_1, \dots, x_k) \in \mathbb{Z}[x_1, \dots, x_k]$. We call f *well-defined k -variate* if

1. $f(0, \dots, 0) \neq 0$.
2. The set of all integer linear combinations of points in $A(f)$ has finite index in $\mathbb{Z}^k$, denoted by $\text{ind } A(f)$.

Remark 2. The set of all integer linear combinations of points in $A(f)$ is the lattice generated by $A(f)$, and $\text{ind } A(f)$ is its determinant.

Theorem 1. For any well-defined k -variate f , the asymptotic bound obtained by the basic strategy is, for any arbitrarily small constant $\varepsilon > 0$:

$$\prod_{j=1}^k X_j^{\int_{N(f)} x_j dV} < W^{\int_{N(f)} \frac{1}{k+1} dV - \varepsilon}.$$

Proof. Using lexicographic order on monomials, we arrange the basis so that the matrix is upper triangular. The diagonal elements of rows corresponding to g are those containing the constant term of $\tilde{f}$. Hence, the diagonal entries are $\prod_{j=1}^k X_j^{d_j(m-1)}$ for the g polynomials, and $W \prod_{j=1}^k X_j^{d_j(m-1)+i_j}$ for the $\tilde{g}$ polynomials. Therefore,

$$\prod_{j=1}^k X_j^{p_j} < W^{p_W}, \quad \text{for } p_j = \sum_{x_1^{i_1} \dots x_k^{i_k} \in M \setminus S} i_j, \quad \text{and } p_W = |S| + 2 - k.$$

Since $p_W = |S| + 2 - k = |(m-1)A(f)| + 2 - k$, Lemma 4 gives

$$\lim_{m \rightarrow \infty} \frac{p_W}{m^k} = \frac{\int_{N(f)} 1 dV}{\text{ind } A(f)}.$$

For p_{j_0} with $j_0 \in \{1, \dots, k\}$,

$$\begin{aligned} p_{j_0} &= \sum_{x_1^{i_1} \dots x_k^{i_k} \in M \setminus S} i_{j_0} \\ &= \sum_{(i_1, \dots, i_k) \in mA(f)} i_{j_0} - \sum_{(i_1, \dots, i_k) \in (m-1)A(f)} i_{j_0}. \end{aligned}$$

By Lemma 5, there exists M and a degree- $(k+1)$ polynomial $g_{j_0}(x)$ such that for all $m > M$,

$$\begin{aligned} \sum_{(i_1, \dots, i_k) \in mA(f)} i_{j_0} &= g_{j_0}(m), \\ \sum_{(i_1, \dots, i_k) \in (m-1)A(f)} i_{j_0} &= g_{j_0}(m-1), \end{aligned}$$

and the leading coefficient of g_{j_0} is $\int_{N(f)} x_{j_0} dV / \text{ind } A(f)$. Hence,

$$\lim_{m \rightarrow \infty} \frac{p_{j_0}}{m^k} = (k+1) \frac{\int_{N(f)} x_{j_0} dV}{\text{ind } A(f)}.$$

Combining these limits with the determinant inequality yields the claimed asymptotic bound after absorbing lower-order terms into ε . $\square$

ε -term. The ε -term is important in Coppersmith's method, but is sometimes overlooked. For univariate modular polynomials, Coppersmith [13] showed that one can efficiently eliminate the ε -term via an elementary brute-force search. However, for k -variate modular polynomials ($k > 1$), Meers and Nowakowski showed in [40, Appendix D] that the ε -term is inherent and cannot be easily eliminated. They showed that this ε -term translates to a sub-exponential (but super-polynomial) search that cannot be efficiently removed by brute force. Note that the ε -term depends on the lattice-construction strategy; the following result applies to the strategy of [40].

ε -term in modular polynomials. To eliminate the ε -term for modular polynomials with modulus M , one must perform a brute-force search with time complexity $2^{\Theta(\log(M)^{1-1/k})}$.

For modular polynomials, the scaling from the LLL algorithm yields a factor $2^{\Theta(m^{k-1})}$, and when we keep only the leading term for asymptotic bounds we miss a factor $M^{\Theta(1/m)}$. Multiplying them together gives

$$2^{\Theta(m^{k-1})} M^{\Theta(1/m)} \geq 2^{\Theta(\log(M)^{1-1/k})},$$

which means that a brute-force search of size $2^{\Theta(\log(M)^{1-1/k})}$ is required to eliminate the ε -term.

We now give an analogous discussion for eliminating the ε -term in integer polynomials.

ε -term in integer polynomials. To eliminate the ε -term for integer polynomials with $W = \|f(x_1 X_1, \dots, x_k X_k)\|_\infty$, one must perform a brute-force search with time complexity $2^{\Theta(\log(W)^{1-\frac{1}{k+1}})}$.

The difference from ε -term in modular polynomials and ε -term in integer polynomials is, for integer polynomials, the scaling from the LLL algorithm yields

$2^{\Theta(m^k)}$, and when keeping only the leading term for asymptotic bounds we miss a factor $W^{\Theta(1/m)}$. Multiplying them together gives

$$2^{\Theta(m^k)} W^{\Theta(1/m)} = 2^{\Theta(m^k) + \Theta(\frac{\log W}{m})} \geq 2^{\Theta(\log(W)^{1 - \frac{1}{k+1}})},$$

which implies a brute-force search of size $2^{\Theta(\log(W)^{1 - \frac{1}{k+1}})}$ is required to eliminate the ε -term.

Remark 3. *We give a rigorous proof of eliminating the ε -term for the lattice construction used in our paper later in Lemma 8. As noted, the ε -term depends on the chosen strategy. Lemma 8 applies to all strategies used in this paper. It does not apply to certain bivariate strategies such as [13, 17], which do not require sub-exponential brute-force search to eliminate the ε -term.*

For many polynomials, it is advantageous to use extra shifts of a certain variable; this is the Jochemsz–May Extended Strategy [31]. We also provide the associated bound formulas for Extended Strategy in Appendix A. And we also generalize our strategy from a single polynomial equation to a system of polynomial equations and provide the associated bound formulas; see Appendix B.

3.2 For k -variable integer equations with total degree δ

In this section, we study k -variable integer equations of total degree δ . The asymptotic bounds above require the input polynomials to be *well-defined k -variate*. When $k = 2$, the well-known bound $X_1 X_2 < W^{1/2 - \varepsilon}$ with $W = \|f(x_1 X_1, x_2 X_2)\|_\infty$ is already available [13, 15–17] without the *well-defined k -variate* condition. Our idea is to consider the family of polynomial equations of total degree δ ; some instances admit better bounds, but if we consider the worst case in this family, we can remove the *well-defined k -variate* condition by slightly modifying the lattice construction. Concretely, we choose a set A larger than $A(f)$ and obtain bounds on small roots for k -variable integer equations of total degree δ . Choosing a larger $A \subseteq \mathbb{Z}^k$ may worsen the bound for some f , but it remains suitable in the worst case for the degree- δ family. We give a detailed discussion below.

Theorem 2. *Fix k and $\delta > 1$. Let $f \in \mathbb{Z}[x_1, \dots, x_k]$ be an irreducible polynomial of total degree δ . Let $X_i \in \mathbb{Z}_{>0}$ for $i = 1, \dots, k$ be upper bounds on the desired integer solution $(x_1^{(0)}, \dots, x_k^{(0)})$, i.e., $|x_i^{(0)}| < X_i$, and let $W = \|f(x_1 X_1, \dots, x_k X_k)\|_\infty$. For $k > 2$, assume Assumption 1 holds. If for some constant $\varepsilon > 0$,*

$$X_1 \cdot \dots \cdot X_k < W^{\frac{1}{\delta} - \varepsilon},$$

then in time $\text{polylog } W$, one can find all integer roots $(x_1^{(0)}, \dots, x_k^{(0)})$ such that $f(x_1^{(0)}, \dots, x_k^{(0)}) = 0$ and $|x_i^{(0)}| \leq X_i$ for $i = 1, \dots, k$.

Proof. We slightly modify the previous strategy. Define

$$\bar{A} = \{(i_1, \dots, i_k) \mid i_j \geq 0 \text{ for all } j, \sum_{j=1}^k i_j \leq \delta\}.$$

and let $\bar{N}$ be the convex hull of $\bar{A}$.

The idea is to replace $A(f)$ by $\bar{A}$ as follows. Let $m \in \mathbb{N}$ and define

$$\begin{aligned} S &= \{x_1^{i_1} \cdots x_k^{i_k} \mid (i_1, \dots, i_k) \in (m-1)\bar{A}\}, \\ M &= \{x_1^{i_1} \cdots x_k^{i_k} \mid (i_1, \dots, i_k) \in m\bar{A}\}. \end{aligned}$$

Set $R = W \prod_{j=1}^k X_j^{l_j}$, where l_j is the largest exponent of x_j among the monomials in S , and define the shift polynomials

$$\begin{aligned} g &= x_1^{i_1} \cdots x_k^{i_k} \tilde{f}(x_1, \dots, x_k) \prod_{j=1}^k X_j^{l_j - i_j} \quad \text{for } x_1^{i_1} \cdots x_k^{i_k} \in S, \\ \tilde{g} &= x_1^{i_1} \cdots x_k^{i_k} R \quad \text{for } x_1^{i_1} \cdots x_k^{i_k} \in M \setminus S, \end{aligned}$$

where $\tilde{f} = a_0^{-1} f \bmod R$ and a_0 is the constant term of f . As before, the diagonal entries are $\prod_{j=1}^k X_j^{d_j(m-1)}$ for the g polynomials and $W \prod_{j=1}^k X_j^{d_j(m-1)+i_j}$ for the $\tilde{g}$ polynomials, hence

$$\prod_{j=1}^k X_j^{p_j} < W^{p_W}, \quad \text{for } p_j = \sum_{x_1^{i_1} \cdots x_k^{i_k} \in M \setminus S} i_j, \quad \text{and } p_W = |S| + 2 - k.$$

Similarly to the proof of Theorem 1, we obtain

$$\begin{cases} \lim_{m \rightarrow \infty} \frac{p_j}{m^k} = \frac{(k+1) \int_{\bar{N}} x_j dV}{\text{ind } \bar{A}} & \text{for } j = 1, \dots, k, \\ \lim_{m \rightarrow \infty} \frac{p_W}{m^k} = \frac{\int_{\bar{N}} 1 dV}{\text{ind } \bar{A}}. \end{cases}$$

Since $x_j \in \bar{A}$ for $j = 1, \dots, k$, we have $\text{ind } \bar{A} = 1$. Furthermore, we have

$$\begin{aligned} \int_{\bar{N}} 1 dV &= \frac{\delta^k}{k!}, \\ \int_{\bar{N}} x_{j_0} dV &= \frac{\delta^{k+1}}{(k+1)!}. \end{aligned}$$

Thus, the asymptotic bound (ignoring the ε -term) is

$$\prod_{j=1}^k X_j^{\frac{(k+1)\delta^{k+1}}{(k+1)!}} < W^{\frac{\delta^k}{k!}}.$$

Rearranging the exponents, this implies (after adding the ε -term) that

$$X_1 \cdots X_k < W^{\frac{1}{\delta} - \varepsilon}.$$

□

As discussed in Section 3.1, removing the ε -term requires sub-exponential time. We now give a rigorous proof for Eliminating the ε -term in Theorem 2.

Lemma 8. *Eliminating the ε -term in Theorem 2 requires time $\exp\left(\log^{\frac{k}{k+1}} W\right)$.*

Proof. The ε -term arises from two parts. First, the analysis of the LLL algorithm introduces a factor involving a power of 2. Second, when we keep only the leading term in the asymptotic estimate of the determinant, we omit a multiplicative factor of the form $W^{\Theta(1/m)}$. To show this, it is necessary to compute at least the second-highest coefficient, rather than only the leading coefficient, in the asymptotic expansion.

A detailed proof is given in Appendix C.1. $\square$

Discussion on $\bar{A}$ versus $A(f)$. In this section, we consider the entire family of polynomials of total degree δ , and therefore we must target the worst-case scenario. For instance, when $\delta = 2$ and $k = 2$, Theorem 2 yields the bound

$$X_1 X_2 < W^{\frac{1}{2} - \varepsilon}.$$

However, for certain specific polynomials f , using the set $A(f)$ leads to a stronger asymptotic bound. For example, when $\delta = 2$ and

$$A(f) = \{(0, 0), (0, 1), (1, 0), (1, 1)\},$$

Theorem 1 gives

$$X_1 X_2 < W^{\frac{2}{3} - \varepsilon}.$$

4 Solving Quadratic Forms with Coppersmith's Method

In this section, we give a success condition under which one can solve the target quadratic form equation $f_{\mathbf{G}}(x_1, \dots, x_k) = \mathbf{x}^t \mathbf{G} \mathbf{x} - d$ in polynomial time. Our approach consists of three steps. In Section 4.1, we generalize the idea of [10] and show that if $\mathbf{G} = \mathbf{B}^t \mathbf{B}$ is LLL-reduced, then one obtains an upper bound on the product of the coordinates of each root. Next, in Section 4.2, we propose a variant of Coppersmith's method for solving polynomial equations when the product of the coordinates of the target root is small. Finally, in Section 4.3, we combine the above ingredients to solve this class of quadratic polynomial equations.

4.1 Upper Bounds for Target Roots

For a matrix $\mathbf{B} \in \mathbb{Z}^{k \times k}$, define its Gram matrix as $\mathbf{G} := \mathbf{B}^t \mathbf{B}$. We study quadratic forms with Gram matrix $\mathbf{G}$. Consider $f_{\mathbf{G}} = \mathbf{x}^t \mathbf{G} \mathbf{x} - d \in \mathbb{Z}[x_1, \dots, x_k]$, which has total degree 2. We can apply Theorem 2 directly to $f_{\mathbf{G}}(\mathbf{x})$. Moreover, we also consider root sizes scaled by c by performing a brute-force search. This search increases the time complexity by a factor of c .

Lemma 9. *Let $\mathbf{B} \in \mathbb{Z}^{k \times k}$ be a full-rank matrix with Gram matrix $\mathbf{G} = \mathbf{B}^t \mathbf{B}$. Let $d > 0$, and suppose the polynomial $f_{\mathbf{G}} = \mathbf{x}^t \mathbf{G} \mathbf{x} - d \in \mathbb{Z}[x_1, \dots, x_k]$ is irreducible. Let $X_i \in \mathbb{Z}_{>0}$ for $i = 1, \dots, k$ be upper bounds on the desired integer solution $(x_1^{(0)}, \dots, x_k^{(0)})$, i.e., $|x_i^{(0)}| < X_i$, and let $W = \|f(x_1 X_1, \dots, x_k X_k)\|_{\infty}$. If there exist constants $\varepsilon > 0$ and integer $c \geq 1$ such that*

$$X_1 \cdots X_k < c \cdot d^{\frac{1}{2} - \varepsilon},$$

then under Assumption 1, in time $c \cdot \text{polylog } W$, one can find all integer roots $(x_1^{(0)}, \dots, x_k^{(0)})$ satisfying

$$f_{\mathbf{G}}(x_1^{(0)}, \dots, x_k^{(0)}) = 0, \quad |x_i^{(0)}| \leq X_i, \quad i = 1, \dots, k.$$

Proof. The idea is to brute-force the $\log_2 c$ most significant bits (MSBs) of some x_{j_0} . After this, we obtain a new upper bound for x_{j_0} , which is the previous upper bound divided by c , i.e., X_{j_0}/c . This step requires time proportional to c , which explains why the total running time is multiplied by the factor c .

A detailed proof can be found in Appendix C.2. $\square$

Lemma 9 requires an upper bound X_i on each variable $|x_i^{(0)}|$. When $\mathbf{B}$ is LLL-reduced, it was shown in [10, Lemma 4.2] that $|x_i^{(0)}|$ can be bounded in terms of d and the Euclidean norm of the i -th column $\mathbf{b}_i$ of $\mathbf{B}$. For the d -Isogeny problem, we expect the basis vectors to have norms of roughly the same magnitude. However, in exceptional cases, one basis vector may be much larger than the others. Without additional assumptions, the resulting bound $X_1 \cdots X_k$ on the product $\prod_i |x_i^{(0)}|$ would not be tight. Nevertheless, the product of the norms of the basis vectors is intrinsically linked to the determinant of $\mathbf{G}$. This relation allows us to derive the following tighter bound on $\prod_i |x_i^{(0)}|$.

Lemma 10. *Given $d > 0$, an LLL-reduced full-rank matrix $\mathbf{B} \in \mathbb{Z}^{k \times k}$, and its Gram matrix $\mathbf{G} = \mathbf{B}^t \mathbf{B}$, if $\mathbf{x}^t \mathbf{G} \mathbf{x} = d$ has a solution $(x_1^{(0)}, \dots, x_k^{(0)})$, then*

$$\prod_{j=1}^k |x_j^{(0)}| \leq \prod_{i=1}^k \frac{\sqrt{d} \cdot \prod_{i \neq j} \|\mathbf{b}_i\|_2}{\det(\mathbf{B})} \leq 2^{\frac{k(k-1)^2}{4}} \sqrt{\frac{d^k}{\det(\mathbf{G})}},$$

where $\mathbf{b}_j$ is the j -th column of $\mathbf{B}$.

Proof. Given linearly independent vectors $\mathbf{b}_1, \dots, \mathbf{b}_k \in \mathbb{R}^k$, their Gram-Schmidt orthogonalization is $\mathbf{b}_i^* = \mathbf{b}_i - \sum_{j < i} \mu_{i,j} \mathbf{b}_j^*$, where $\mu_{i,j} = \frac{\langle \mathbf{b}_i, \mathbf{b}_j^* \rangle}{\langle \mathbf{b}_j^*, \mathbf{b}_j^* \rangle}$. As the matrix $\mathbf{B}$ is LLL-reduced, then we have

$$\|\mathbf{b}_i^*\|_2^2 \geq \frac{1}{2} \|\mathbf{b}_{i-1}^*\|_2^2.$$

Hence, for any $i_0 \leq j_0$,

$$\|\mathbf{b}_{i_0}^*\|_2^2 \leq 2^{j_0 - i_0} \|\mathbf{b}_{j_0}^*\|_2^2.$$

Therefore, for any j_0 ,

$$\begin{aligned}\|\mathbf{b}_{j_0}\|_2^2 &= \|\mathbf{b}_{j_0}^*\|_2^2 + \sum_{j=1}^{j_0-1} \mu_{j_0,j}^2 \|\mathbf{b}_j^*\|_2^2 \\ &\leq \left(1 + \frac{1}{4} \sum_{j=1}^{j_0-1} 2^{j_0-j}\right) \|\mathbf{b}_{j_0}^*\|_2^2 = \frac{2^{j_0+2}}{4} \|\mathbf{b}_{j_0}^*\|_2^2 \leq 2^{j_0-1} \|\mathbf{b}_{j_0}^*\|_2^2.\end{aligned}$$

Since $\mathbf{G} = \mathbf{B}^t \mathbf{B}$, any integer root $\mathbf{x}$ of

$$\mathbf{x}^t \mathbf{G} \mathbf{x} = d$$

satisfies

$$\mathbf{x}^t \mathbf{B}^t \mathbf{B} \mathbf{x} = (\mathbf{B} \mathbf{x})^t (\mathbf{B} \mathbf{x}) = \|\mathbf{v}\|_2^2 = d,$$

where we define $\mathbf{v} = \mathbf{B} \mathbf{x}$. Thus $\|\mathbf{v}\|_2 = \sqrt{d}$.

By Cramer's rule, for any index j_0 ,

$$|x_{j_0}^{(0)}| = \frac{\det(\mathbf{B}_{j_0})}{\det(\mathbf{B})} \leq \frac{\|\mathbf{v}\|_2 \cdot \prod_{j \neq j_0} \|\mathbf{b}_j\|_2}{\det(\mathbf{B})}, \quad (4)$$

where $\mathbf{B}_{j_0}$ is obtained from $\mathbf{B}$ by replacing its j_0 -th column with $\mathbf{v}$, and $\mathbf{b}_j$ denotes the j -th column of $\mathbf{B}$.

Moreover,

$$\det(\mathbf{G}) = \det(\mathbf{B}^t \mathbf{B}) = \det(\mathbf{B})^2 = \prod_{j=1}^k \|\mathbf{b}_j^*\|_2^2,$$

where $\mathbf{b}_j^*$ denotes the j -th Gram-Schmidt vector of $\mathbf{B}$.

We have

$$\prod_{j=1}^k \|\mathbf{b}_j\|_2 \leq \left(\prod_{j=1}^k 2^{j-1}\right)^{\frac{1}{2}} \prod_{j=1}^k \|\mathbf{b}_j^*\|_2 = 2^{\frac{k(k-1)}{4}} \prod_{j=1}^k \|\mathbf{b}_j^*\|_2 = 2^{\frac{k(k-1)}{4}} \det(\mathbf{B}). \quad (5)$$

Multiplying the bounds for all j_0 yields

$$\begin{aligned}\prod_{j=1}^k |x_j^{(0)}| &\leq \prod_{i=1}^k \frac{\|\mathbf{v}\|_2 \cdot \prod_{i \neq j} \|\mathbf{b}_i\|_2}{\det(\mathbf{B})} \\ &= \frac{\|\mathbf{v}\|_2^k \cdot \left(\prod_{j=1}^k \|\mathbf{b}_j\|_2\right)^{k-1}}{\det(\mathbf{B})^k} \\ &\leq \frac{d^{\frac{k}{2}} \cdot 2^{\frac{k(k-1)^2}{4}} \cdot \left(\prod_{j=1}^k \|\mathbf{b}_j^*\|_2\right)^{k-1}}{\left(\prod_{j=1}^k \|\mathbf{b}_j^*\|_2\right)^k} \\ &= 2^{\frac{k(k-1)^2}{4}} \sqrt{\frac{d^k}{\det(\mathbf{G})}}.\end{aligned}$$

□

By Equation (4) and Equation (5), we can set $B_j = 2^{\frac{k(k-1)}{4}} \sqrt{\frac{d}{\|\mathbf{b}_j\|_2^2}}$ such that each root $(x_1^{(0)}, \dots, x_k^{(0)})$ satisfies

$$|x_j^{(0)}| \leq B_j = 2^{\frac{k(k-1)}{4}} \sqrt{\frac{d}{\|\mathbf{b}_j\|_2^2}},$$

where $\mathbf{b}_j$ denotes the j -th column of $\mathbf{B}$. Focusing on the product of the coordinates leads to a slightly tighter bound:

$$\left| \prod_{j=1}^k x_j^{(0)} \right| < 2^{\frac{k(k-1)^2}{4}} \sqrt{\frac{d^k}{\det(\mathbf{G})}} < \prod_{j=1}^k B_j,$$

which motivate the following variant in Section 4.2.

4.2 A Variant of Coppersmith's Method.

Here we show that one can find all roots satisfying

$$\prod_{j=1}^k (|x_j^{(0)}| + 1) < d^{1/2-\varepsilon},$$

rather than only those roots constrained by individual bounds $|x_j^{(0)}| < B_j$ for given B_j , provided that $\prod_{j=1}^k \lceil B_j \rceil < d^{1/2-\varepsilon}$ for some constant $\varepsilon > 0$. The procedure is as follows. Define

$$\mathcal{D} = \left\{ (d_1, \dots, d_k) \left| \sum_{i=1}^k d_i = \lceil \log_2(d^{1/2-\varepsilon}) \rceil + k \right. \right\} \subseteq \mathbb{Z}^k.$$

For each tuple $(d_1, \dots, d_k) \in \mathcal{D}$, set $X_i = 2^{d_i}$ and run Coppersmith's method with these bounds. Crucially, $|\mathcal{D}|$ is polynomial in the input size, so the overall running time remains polynomial (see Theorem 3).

Furthermore, we obtain the following corollary: all solutions satisfying an upper bound B on the product $\prod_{i=1}^k |x_i^{(0)}|$, with the additional condition that each variable is non-zero, can be found efficiently (see Corollary 1). Indeed, when $|x_i^{(0)}| \geq 1$ for all i , we have $|x_i^{(0)}| + 1 \leq 2|x_i^{(0)}|$, so

$$\prod_{j=1}^k (|x_j^{(0)}| + 1) \leq 2^k \left| \prod_{j=1}^k x_j^{(0)} \right|.$$

Although this variant does not improve the asymptotic bounds required for the results in this paper, it is of independent interest. In particular, it may be useful in scenarios where the product bound is much smaller than the product of the individual coordinate bounds.

Theorem 3. *Given $d > 0$ and a positive-definite matrix $\mathbf{G} \in \mathbb{Z}^{k \times k}$ for fixed k , suppose $f_{\mathbf{G}} = \mathbf{x}^t \mathbf{G} \mathbf{x} - d \in \mathbb{Z}[x_1, \dots, x_k]$ is irreducible. If there exist constants $\varepsilon > 0$ and an integer $1 \leq c \leq d$ such that*

$$\prod_{j=1}^k (|x_j^{(0)}| + 1) < c d^{\frac{1}{2} - \varepsilon},$$

then one can find all such solutions in time $c \cdot \text{polylog}(d, \|\mathbf{G}\|_{\infty})$ under Assumption 1.

We prove this theorem in four steps: first, we construct a set of upper bounds that covers every solution of interest; next, we show that every valid solution is captured by one of these bounds; then, we use Lemma 9 to enumerate solutions within these bounds; and finally, we bound the running time. The proof appears in Appendix C.3.

Moreover, we obtain the following corollary: all solutions for which an upper bound on the product of variables is small and each variable is non-zero can also be found efficiently.

Corollary 1. *Given $d > 0$ and a positive-definite matrix $\mathbf{G} \in \mathbb{Z}^{k \times k}$ for fixed k , suppose $f_{\mathbf{G}} = \mathbf{x}^t \mathbf{G} \mathbf{x} - d$ is irreducible. If there exist constant $\varepsilon > 0$ and an integer $1 \leq c \leq d/2^k$ such that*

$$|x_1^{(0)} \cdots x_k^{(0)}| < c d^{\frac{1}{2} - \varepsilon} \quad \text{and} \quad \prod_{i=1}^k x_i^{(0)} \neq 0,$$

then all such solutions can be found in time $c \cdot \text{polylog}(d, \|\mathbf{G}\|_{\infty})$ under Assumption 1.

Proof. Since

$$|x_1^{(0)} \cdots x_k^{(0)}| < c d^{\frac{1}{2} - \varepsilon} \quad \text{and} \quad \prod_{j=1}^k x_j^{(0)} \neq 0,$$

we have

$$\prod_{j=1}^k (|x_j^{(0)}| + 1) < 2^k |x_1^{(0)} \cdots x_k^{(0)}| < c 2^k d^{\frac{1}{2} - \varepsilon}.$$

Hence, by Theorem 3, the solutions can be found in time $c \cdot 2^k \cdot \text{polylog}(d, \|\mathbf{G}\|_{\infty}) = c \cdot \text{polylog}(d, \|\mathbf{G}\|_{\infty})$, as claimed. $\square$

Remark 4. *One may wonder whether Coppersmith's method can be applied when only a bound on the product $|\prod_i x_i^{(0)}| < d^{1/2 - \varepsilon}$ is available. Unfortunately, this appears difficult to achieve. Assume, for contradiction, that one could efficiently recover all roots $(x_1^{(0)}, \dots, x_k^{(0)})$ satisfying $|\prod_{i=1}^k x_i^{(0)}| < d^{1/2 - \varepsilon}$. Consider the case $k = 3$ with*

$$g(x_1, x_2) = x_1^2 + x_2^2 - N, \quad f(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2 - N,$$

where $N = pq$ is an RSA modulus with $p \equiv q \equiv 1 \pmod{4}$. Every root $(x_1^{(0)}, x_2^{(0)})$ of g yields a root $(x_1^{(0)}, x_2^{(0)}, 0)$ of f , and clearly $x_1^{(0)} x_2^{(0)} \cdot 0 < N^{1/2-\varepsilon}$. By assumption, we could therefore recover all roots of g , i.e., all representations of N as a sum of two squares. By Jacobi's two-square theorem [27], such an N admits at least two distinct representations $N = x_1'^2 + x_2'^2 = x_1''^2 + x_2''^2$ with $(x_1', x_2') \neq (x_1'', x_2'')$. In $\mathbb{Z}[i]$ this gives $N = (x_1' + x_2'i)(x_1' - x_2'i) = (x_1'' + x_2''i)(x_1'' - x_2''i)$. Let $w := \gcd(x_1' + x_2'i, x_1'' + x_2''i) = a + bi$ be the Gaussian integer gcd. Since w divides $x_1' + x_2'i$, we have $a^2 + b^2 \mid N$. Thus $a^2 + b^2$ is a non-trivial factor of N , yielding a factorization of the RSA modulus [30]. This contradicts the hardness of factoring.

4.3 Solving Quadratic Forms on Gram Matrix

Suppose $\mathbf{B}$ is LLL reduced. In Lemma 10, we established that every integer root of $f_{\mathbf{G}}$ satisfies an upper bound on the product of its coordinates. However, Theorem 3 requires an upper bound on the quantity $\prod_{j=1}^k (|x_j| + 1)$ instead of merely $\prod_{j=1}^k |x_j|$. In the following, we show that when d is sufficiently large, Lemma 10 implies that we can apply Theorem 3 to recover all the integral roots of $f_{\mathbf{G}}$.

Theorem 4. *Let $\mathbf{B} \in \mathbb{Z}^{k \times k}$ be full-rank with Gram matrix $\mathbf{G} = \mathbf{B}^t \mathbf{B}$. Assume that $f_{\mathbf{G}}(\mathbf{x}) = \mathbf{x}^t \mathbf{G} \mathbf{x} - d$ is irreducible. Suppose there exists an integer $c_1 \geq 1$ such that*

$$c_1 d > \max_j \{ \|\mathbf{b}_j^{\text{LLL}}\|_2^2 \},$$

where $\mathbf{b}_j^{\text{LLL}}$ is the j -th column of the LLL-reduced basis of $\mathbf{B}$. Furthermore, suppose there exists an integer $c_2 \geq 1$ and a constant $\varepsilon > 0$ such that

$$c_2 \det(\mathbf{G}) > d^{k-1+\varepsilon}.$$

Then, under Assumption 1, all integer roots $\mathbf{x} \in \mathbb{Z}^k$ of $f_{\mathbf{G}}$ can be found in time

$$\sqrt{c_1^k c_2} \cdot \text{polylog}(d, \|\mathbf{G}\|_{\infty}).$$

Proof. Run LLL on $\mathbf{B}$. Then there exists a unimodular matrix $\mathbf{U}$ such that $\mathbf{BU}$ is LLL-reduced. Since

$$\det(\mathbf{U}^t \mathbf{G} \mathbf{U}) = \det(\mathbf{G}),$$

and irreducibility of $f_{\mathbf{G}}$ carries over to $f_{\mathbf{U}^t \mathbf{G} \mathbf{U}}$, we may assume without loss of generality that $\mathbf{B}$ is already LLL-reduced.

Let $\mathbf{x}$ be an integer root of

$$\mathbf{x}^t \mathbf{G} \mathbf{x} = d.$$

Since $\mathbf{G} = \mathbf{B}^t \mathbf{B}$, we have

$$\mathbf{x}^t \mathbf{B}^t \mathbf{B} \mathbf{x} = \|\mathbf{B} \mathbf{x}\|_2^2 = d.$$

Define $\mathbf{v} = \mathbf{B}\mathbf{x}$; then $\|\mathbf{v}\|_2 = \sqrt{d}$.

By Equation (4), for each j_0 ,

$$|x_{j_0}^{(0)}| \leq \frac{\|\mathbf{v}\|_2 \cdot \prod_{j \neq j_0} \|\mathbf{b}_j\|_2}{\det(\mathbf{B})} = \frac{\sqrt{d} \prod_{j \neq j_0} \|\mathbf{b}_j\|_2}{\det(\mathbf{B})}.$$

Using $\sqrt{c_1 d} / \|\mathbf{b}_{j_0}\|_2 > 1$, we get

$$\frac{\sqrt{c_1 d} \prod_{j \neq j_0} \|\mathbf{b}_j\|_2}{\det(\mathbf{B})} \geq \frac{\sqrt{c_1 d}}{\|\mathbf{b}_{j_0}\|_2} \geq 1.$$

Hence

$$\prod_{j=1}^k (|x_j^{(0)}| + 1) \leq \prod_{j=1}^k \left(\frac{\sqrt{d} \prod_{i \neq j} \|\mathbf{b}_i\|_2}{\det(\mathbf{B})} + 1 \right) \leq \prod_{j=1}^k \left(\frac{\sqrt{c_1 d} \prod_{i \neq j} \|\mathbf{b}_i\|_2}{\det(\mathbf{B})} + \frac{\sqrt{c_1 d} \prod_{i \neq j} \|\mathbf{b}_i\|_2}{\det(\mathbf{B})} \right),$$

and by Lemma 10 this becomes

$$\prod_{j=1}^k (|x_j^{(0)}| + 1) \leq 2^k \prod_{j=1}^k \frac{\sqrt{c_1 d} \prod_{i \neq j} \|\mathbf{b}_i\|_2}{\det(\mathbf{B})} \leq 2^{\frac{k(k-1)^2}{4} + k} \sqrt{\frac{c_1^k d^k}{\det(\mathbf{G})}}.$$

Using $c_2 \det(\mathbf{G}) > d^{k-1+\varepsilon}$, we obtain

$$\prod_{j=1}^k (|x_j^{(0)}| + 1) < 2^{\frac{k^3 - 2k^2 + 5k}{4}} c_1^{k/2} c_2^{1/2} d^{\frac{1}{2} - \frac{\varepsilon}{2}}.$$

Thus setting

$$c = 2^{\frac{k^3 - 2k^2 + 5k}{4}} \sqrt{c_1^k c_2}$$

in Theorem 3 gives the claimed running time. $\square$

Remark 5. *Eliminating the ε -term requires a brute-force factor $2^{\Theta(\log^{\frac{k}{k+1}} d)}$ by Lemma 8.*

Remark 6. *When $f_{\mathbf{G}}$ is reducible, there are only two cases:*

1. *Case $\gcd(\mathbf{G}, d) \neq 1$: In this situation, the bound becomes weaker, since now $W := \|f_{\mathbf{G}}(x_1 X_1, \dots, x_k X_k)\|_{\infty} / \gcd(\mathbf{G}, d)$.*
2. *Case $\mathbf{G}$ is not full rank: Here $f_{\mathbf{G}}$ can be factored as a product of two polynomials with degree 1. In this case, finding a solution reduces to solving the Closest Vector Problem (CVP) in a k -dimensional lattice. For fixed k , CVP can be solved in polynomial time, which implies that the bound still holds.*

5 Finding fixed-degree isogenies

5.1 Applying four-variable Coppersmith's method

In this section, we apply our explicit formulas for four-variable Coppersmith's method to the fixed-degree isogeny problem. We are given supersingular curves E_1 and E_2 and an integer d , and our goal is to find some isogeny $\varphi : E_1 \rightarrow E_2$ of degree d . We follow the blueprint of Benčina et al. [10], citing the algorithms used.

1. Compute $\text{End}(E_1)$ and $\text{End}(E_2)$ [44, Theorem 8.8].
2. Compute maximal orders $\mathcal{O}_1 \cong \text{End}(E_1)$ and $\mathcal{O}_2 \cong \text{End}(E_2)$ in $B_{p,\infty}$ [41, Proposition 6.2].
3. Compute a connecting ideal I between $\mathcal{O}_1$ and $\mathcal{O}_2$ [33, Algorithm 3.5].
4. Compute a Minkowski-reduced basis $(\sigma_1, \sigma_2, \sigma_3, \sigma_4)$ of I using LLL.
5. Let $\mathbf{G} = (\frac{1}{2 \text{nr}(I)} \text{trd}(\sigma_i \overline{\sigma_j}))_{i,j} \in \mathbb{Z}^{4 \times 4}$ be the Gram matrix with respect to the reduced norm of I .
6. (To be specified): Find an integral root of the four-variable polynomial

$$f_{\mathbf{G}}(x_1, \dots, x_4) = \mathbf{x}^t \mathbf{G} \mathbf{x} - d.$$

7. Compute an ideal J equivalent to I of norm d using the root of $f_{\mathbf{G}}$. Let $\alpha = \sum_{i=1}^4 x_i \sigma_i$ and compute $J = I \frac{\overline{\alpha}}{\text{nr}(I)}$ [34, Lemma 5].
8. Translate J to an isogeny $\varphi : E_1 \rightarrow E_2$ of degree d [43].

The first step runs in $\tilde{O}(\sqrt{p})$ time. The remaining specified steps run in polynomial time in $\log p$ and $\log d$. It only remains to choose an algorithm for Step 6 in order to compute a root of $f_{\mathbf{G}}$. Benčina et al. [10] applied Coppersmith's method in two or three variables, as well as Cornacchia's algorithm. Since $f_{\mathbf{G}}$ is a polynomial in four variables, these approaches require guessing one or two of the variables. Instead, we use our new k -variate Coppersmith algorithm from Section 3, which can be applied directly to find roots of $f_{\mathbf{G}}$ without any guessing.

In order to apply Coppersmith's method, we make the same average-case assumption on the successive minima of $\text{Hom}(E_1, E_2)$ as in [10].

Assumption 2. *Let $\mathcal{O}_1, \mathcal{O}_2$ be maximal orders in the quaternion algebra $B_{p,\infty}$, and let I be a connecting ideal of $\mathcal{O}_1$ and $\mathcal{O}_2$. Let $\sigma_1, \sigma_2, \sigma_3, \sigma_4$ be a Minkowski-reduced basis for I (as a $\mathbb{Z}$ -lattice). We assume that $q_I(\sigma_i) = \tilde{O}(\sqrt{p})$ for $i = 1, \dots, 4$.*

By Minkowski's bound for I , we know that the product of the successive minima is $\Theta(p^2)$. We make the assumption that the successive minima are roughly of the same size. When the maximal orders $\mathcal{O}_1$ and $\mathcal{O}_2$ are chosen uniformly at random, experimentally this holds with overwhelming probability, as noted in [10, 34].

There are two reasons why Benčina et al. require this assumption:

1. To derive upper-bounds $|x_i| \leq X_i$ for the roots of $f_{\mathbf{G}}$. When $d = p^{1/2+\delta}$ for some $\delta > 0$, this assumption implies that $|x_i| \leq 8 \cdot p^{\delta/2}$ [10, Corollary 4.4]. They use this to bound the complexity of guessing a variable.

2. To bound the infinity norm of $\mathbf{G}$, to ensure that Coppersmith can be applied for a specific d .

With the four-variable Coppersmith algorithm, we no longer need Assumption 2 to derive individual bounds on the $|x_i|$. However, we still need to bound the infinity norm of $\mathbf{G}$, to ensure that we can apply Theorem 4. We cannot entirely remove this assumption without obtaining a weaker result. Note that is possible to generalize the assumption to $d > \max\{q_I(\sigma_i)\}/c$ for $c = p^{o(1)}$.

Theorem 5. *Let $\mathcal{O}_1, \mathcal{O}_2$ be maximal orders in $B_{p,\infty}$ and let $d = p^{1/2+\delta}$ for some constant $0 < \delta < \frac{1}{6}$. Let I be an ideal connecting $\mathcal{O}_1$ and $\mathcal{O}_2$. Then, under Assumption 1 and Assumption 2, there exists a classical algorithm that computes an element $\alpha \in I$ with $q_I(\alpha) = d$ in time polynomial in $\log p$, or determines that no such element exists.*

Proof. We follow the blueprint, using the new four-variable Coppersmith algorithm to find a root of $f_{\mathbf{G}}$. We will use Theorem 4 to show that a root can be found in time polynomial in $\log p$. To apply the theorem, we first need to verify that all of the necessary conditions are met.

(i) $\mathbf{B}$ has full rank. The columns of the matrix $\mathbf{B}$ is the embedding of the Minkowski-reduced basis $\sigma_1, \dots, \sigma_4$ into $\mathbb{R}^4$ with respect to the reduced norm q_I . Hence, $\mathbf{B}$ has full rank.

(ii) $\log\|\mathbf{G}\|_{\infty} = \text{polylog}(p)$. This follows as

$$\|\mathbf{G}\|_{\infty} = \max_{i,j} \{|\langle \sigma_i, \sigma_j \rangle| / \text{nrd}(I)\} \leq \max_{i,j} \left\{ \sqrt{q_I(\sigma_i)} \sqrt{q_I(\sigma_j)} \right\} \leq q_I(\sigma_4) \leq p^2/4,$$

using Equation (2) for the last inequality.

(iii) $c_2 \det(\mathbf{G}) > d^{3+\varepsilon}$ for some constants $\varepsilon, c_2 > 0$. We have $\det(\mathbf{G}) = p^2/16$. Set $\varepsilon = \frac{1-6\delta}{1+2\delta}$. When $0 < \delta < \frac{1}{6}$, we have $\varepsilon > 0$. By direct computation we see that $d^{3+\varepsilon} < p^2 < 16 \det(\mathbf{G})$.

(iv) There exist a $c_1 = \text{polylog}(p)$ such that $cd > m$, where $m := \max\{\|\mathbf{b}_i\|_2^2\} = \max\{q_I(\sigma_i)\}$. We have $d > \sqrt{p}$. By Assumption 2, we have $m = \tilde{O}(\sqrt{p})$. By definition, there are positive constants a and p_0 and a polynomial $t(x)$ such that for $p \geq p_0$, $m \leq a \cdot \sqrt{p} \cdot t(\log p)$. When $p < p_0$, we can upper bound m by the constant $p_0^2/4$ using Equation (2). By setting $c_1 = \max(a \cdot t(\log p), p_0^2/4)$, we get that $c_1 d > m$ and $c_1 = \text{polylog}(p)$.

(v) The polynomial $f_{\mathbf{G}}$ is irreducible. Suppose for the sake of contradiction that it was not. There are two cases to consider.

First suppose that some positive integer c divides $f_{\mathbf{G}}$. Then c must divide $\mathbf{G}$. Since $I\bar{I} = \text{nrd}(I)\mathcal{O}_2$, it holds that $\text{trd}(I\bar{I}) = 2 \text{nrd}(I)\mathbb{Z}$. At the same time $I\bar{I} = \sum_{i,j} \sigma_i \bar{\sigma}_j \mathbb{Z}$, so $\sum_{i,j} \frac{1}{2} \text{trd}(\sigma_i \bar{\sigma}_j) \mathbb{Z} = \text{nrd}(I)\mathbb{Z}$ and $\sum_{i,j} \frac{1}{2 \text{nrd}(I)} \text{trd}(\sigma_i \bar{\sigma}_j) \mathbb{Z} = \mathbb{Z}$. Hence, the gcd of the entries of $\mathbf{G}$ is 1. We conclude that no positive integer divides neither $\mathbf{G}$ nor $f_{\mathbf{G}}$.

Next, suppose that $f_{\mathbf{G}}$ factors as a product of two linear polynomials

$$f_{\mathbf{G}} = (a_1 x_1 + a_2 x_2 + a_3 x_3 + a_4 x_4 + a_0)(b_1 x_1 + b_2 x_2 + b_3 x_3 + b_4 x_4 + b_0).$$

We have $-d = a_0 b_0 \neq 0$, so $a_0, b_0 \neq 0$. Since $f_{\mathbf{G}}$ contains no monomials of degree 1, $a_0 b_i + a_i b_0 = 0$ for all $1 \leq i \leq 4$. This implies that $a_i/b_i = -a_0/b_0$ if $b_i \neq 0$ and $a_i = 0$ if $b_i = 0$. We get

$$a_1 x_1 + a_2 x_2 + a_3 x_3 + a_4 x_4 + a_0 = -\frac{a_0}{b_0} (b_1 x_1 + b_2 x_2 + b_3 x_3 + b_4 x_4 - b_0).$$

Thus,

$$f_{\mathbf{G}} = -\frac{a_0}{b_0} ((b_1 x_1 + b_2 x_2 + b_3 x_3 + b_4 x_4)^2 - b_0^2)$$

and

$$\mathbf{x}^t \mathbf{G} \mathbf{x} = -\frac{a_0}{b_0} (b_1 x_1 + b_2 x_2 + b_3 x_3 + b_4 x_4)^2 = c (\mathbf{b}^t \mathbf{x})^2 = \mathbf{x}^t (c \mathbf{b} \mathbf{b}^t) \mathbf{x},$$

where $\mathbf{b} = (b_1, b_2, b_3, b_4)$ and $c = -a_0/b_0$. Since $\mathbf{G}$ is symmetric, we have $\mathbf{G} = c \mathbf{b} \mathbf{b}^t$. It follows that every row of G is a rational multiple of $\mathbf{b}$, so the rank of $\mathbf{G}$ is 1. We reach a contradiction, as $\det(\mathbf{G}) \neq 0$. $\square$

Remark 7. Achieving $d = \Theta(p^{2/3})$ requires a brute-force factor $2^{\Theta(\log^{\frac{4}{5}} p)}$ by Lemma 8.

Corollary 2. Let $d < p^{2/3-\varepsilon}$ for some constant $\varepsilon > 0$. Under Assumption 1 and Assumption 2, there is a polynomial time reduction from d -Isogeny to EndRing.

5.2 New bound for smooth degrees

When the degree d is too large to apply Coppersmith directly, but d is smooth, we can apply the hybrid approach of [10]. The idea is to guess parts of the isogeny, then solve for the rest. On a high-level, the hybrid approach works as follows.

1. Decompose the degree as $d = d_{\text{guess}} \cdot d_{\text{solve}}$.
2. Compute $\text{End}(E_1)$ and $\text{End}(E_2)$ [44, Theorem 8.8].
3. Enumerate the set of isogenies $\varphi_{\text{guess}} : E_1 \rightarrow E_{\text{guess}}$ from E_1 of degree d_{guess} .
For each:
 - (a) Compute $\text{End}(E_{\text{guess}})$ by translating $\text{End}(E_1)$ through φ_{guess} .
 - (b) Following the blueprint in Section 5.1, compute an isogeny $\varphi_{\text{solve}} : E_{\text{guess}} \rightarrow E_2$ of degree d_{solve} , or decide that no such isogeny exists.
 - (c) If the previous step succeeds, output the d -isogeny $\varphi = \varphi_{\text{solve}} \circ \varphi_{\text{guess}}$.

By applying our four-variable Coppersmith algorithm to the hybrid approach, we improve on the state-of-the-art for smooth d .

Theorem 6. Let $d = p^{2/3+\delta}$ be a smooth integer, where $\delta > 0$ is a constant. Let E_1, E_2 be d -isogenous supersingular elliptic curves over $\mathbb{F}_{p^2}$. Additionally, let ε be a constant $0 < \varepsilon < \frac{2}{3}$. Then under Assumption 1 and Assumption 2, there exists a classical algorithm for d -Isogeny running in time

$$\tilde{O}(\max\{p^{1/2}, p^{\delta+\varepsilon}\}).$$

Proof. To decompose d , we first compute its prime factorization $d = \prod_{i=1}^k \ell_i^{e_i}$. We label the prime factors so that $\ell_1 < \dots < \ell_k$. To compute d_{solve} , we compute the sequence $\ell_1, \ell_1^2, \dots, \ell_1^{e_1}, \ell_1^{e_1} \ell_2, \dots, \ell_1^{e_1} \ell_2^{e_2}, \dots, d$. We let d_{solve} be the last integer less than $p^{2/3-\varepsilon}$. Since $d > p^{2/3-\varepsilon}$, we must have $d_{\text{solve}} \cdot \ell_k > p^{2/3-\varepsilon}$. It follows that

$$d_{\text{guess}} = d/d_{\text{solve}} < \ell_k d/p^{2/3-\varepsilon} = \tilde{O}(d/p^{2/3-\varepsilon}) = \tilde{O}(p^{\delta+\varepsilon}).$$

Since $d < p^{2/3+\delta}$ is smooth, these steps can be computed in time $\text{polylog}(d) = \text{polylog}(p)$.

Next, we expand on how we enumerate the set of degree d_{guess} isogenies from E_1 . We only enumerate them up to post-composition with an isomorphism. Let $d_{\text{guess}} = \prod_{i=1}^k \ell_i^{f_i}$. We have $\varphi_{\text{guess}} = \varphi_k \circ \dots \circ \varphi_1$, where $\deg(\varphi_i) = \ell_i^{f_i}$.

For each i , we have $\varphi_i = [\ell_i^m] \varphi'_i$ for some m , where φ'_i is cyclic of degree ℓ_i^n with $n + 2m = f_i$. For $n > 0$, there are $\ell_i^{n-1}(\ell_i + 1)$ cyclic isogenies of degree ℓ_i^n , up to isomorphism. There are $\ell_i + 1$ choices for the first ℓ_i -step, then there are ℓ_i non-backtracking choices for each subsequent step. We obtain that the number of $\ell_i^{f_i}$ -isogenies is $O(\ell_i^{f_i})$, and that the number of d_{guess} -isogenies is

$$\prod_{i=1}^k O(\ell_i^{f_i}) = O(d_{\text{guess}}) = \tilde{O}(p^{\delta+\varepsilon}).$$

To enumerate the possible φ_{guess} , we enumerate over the choices for each prime ℓ_i . For a fixed ℓ_i , we range over all integers m and all tuples

$$j_1 \in \{0, \dots, \ell_i\}, \quad j_2, \dots, j_n \in \{0, \dots, \ell_i - 1\},$$

such that $f_i = n + 2m$. From the tuple $(j_1, \dots, j_n)$ we construct φ'_i as the corresponding sequence of ℓ_i -isogenies. At the t -th step, we deterministically sample a basis (P, Q) of the ℓ_i -torsion by computing the ℓ_i -th division polynomial, following [45]. These torsion points lie in an extension field $\mathbb{F}_{p^{2k}}$ with $k = O(\ell_i^2)$. We then choose the kernel of the t -th ℓ_i -isogeny as $\langle P + [j_t]Q \rangle$ for $0 \leq j_t < \ell_i$, and as $\langle Q \rangle$ when $j_1 = \ell_i$. Finally, we compute the ℓ_i -isogeny using Vélú's formulas. Overall, the cost of computing each ℓ_i -isogeny is polynomial in $\log p$ and ℓ_i .

While computing φ_{guess} , we also compute the endomorphism ring of the codomain. Given an ℓ_i -isogeny $\psi : E \rightarrow E'$ and the maximal order $\mathcal{O} \cong \text{End}(E)$, we compute the left $\mathcal{O}$ -ideal I_ψ corresponding to ψ . We do this by adapting [49, Theorem 7.1] to work for the order $\mathcal{O}$, by replacing the ideal-to-isogeny algorithm with Clapoti [43]. Given I_ψ , we compute $\mathcal{O}' := \mathcal{O}_R(I_\psi)$. Overall, we compute $\mathcal{O}' \cong \text{End}(E_{\text{guess}})$ in time polynomial in $\log p$ and ℓ_i .

We are now ready to analyze the runtime of Step 3. We iterate over a set of $O(d_{\text{guess}}) = \tilde{O}(p^{\delta+\varepsilon})$ isogenies. We compute φ_{guess} and $\mathcal{O}_{\text{guess}}$ by a sequence of prime degree isogenies. Since $\max_i \ell_i = \text{polylog } p$, the computation per ℓ_i -isogeny is $\text{polylog } p$. The number of prime degree isogenies we compute are $\sum_i f_i \leq (\frac{2}{3} + \delta) \log p$. We conclude that Step 3 runs in time

$$O(d_{\text{guess}} \cdot \text{polylog}(p)) = \tilde{O}(p^{\delta+\varepsilon}).$$

□

We obtain an analogous result for the quantum setting. We modify the hybrid approach in two ways: 1) we use a quantum algorithm to compute $\text{End}(E_1)$ and $\text{End}(E_2)$ and 2) we apply Grover's algorithm to the enumeration step.

Theorem 7. *Let $d = p^{2/3+\delta}$ be a smooth integer, where $\delta > 0$ is a constant. Let E_1, E_2 be d -isogenous supersingular elliptic curves over $\mathbb{F}_{p^2}$. Additionally, let ε be a constant $0 < \varepsilon < \frac{2}{3}$. Then under Assumption 1 and Assumption 2, there exists a quantum algorithm for d -Isogeny running in time*

$$\tilde{O}(\max\{p^{1/4}, p^{(\delta+\varepsilon)/2}\}).$$

We compare our result with prior work in the classical and quantum settings in Figure 1 and Figure 2, respectively.

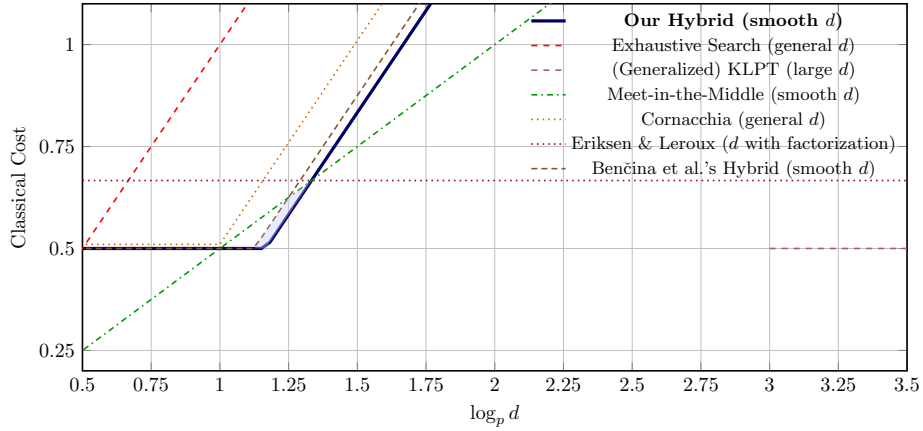


Fig. 1: Classical setting

Remark 8. *As noted in Remark 7, obtaining $\varepsilon = 0$ blows up the runtime by a sub-exponential factor. The runtime of the classical algorithm becomes*

$$\tilde{O}(\max\{p^{1/2}, \exp(O(\log^{4/5} p)) \cdot p^\delta\}).$$

and the runtime of the quantum algorithm becomes

$$\tilde{O}(\max\{p^{1/4}, \exp(O(\log^{4/5} p)) \cdot p^{\delta/2}\}).$$

Remark 9. *We also present results based on the bivariate Coppersmith method [15], which does not rely on Assumption 1 and incurs no sub-exponential terms. Specifically, we obtain two results:*

1. *For general large d , we solve d -Isogeny, improving upon exhaustive search in the regime $p < d < p^2$, though still weaker than the Cornacchia-based method of [10].*

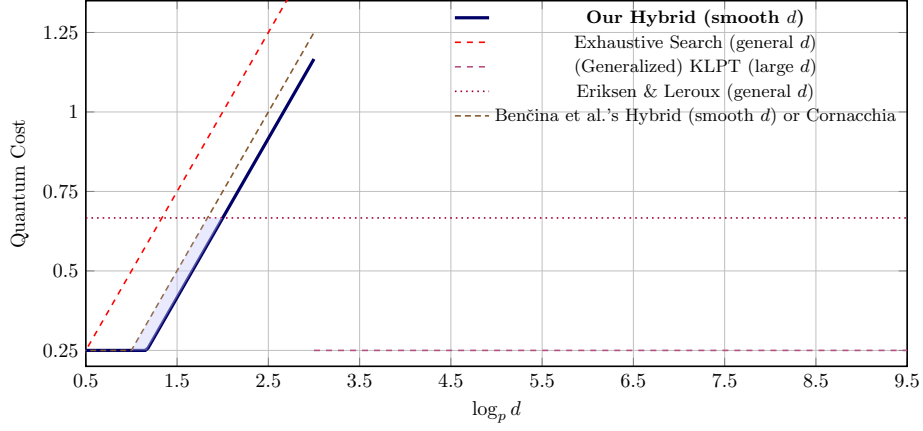


Fig. 2: Quantum setting

2. For smooth d in the quantum setting, a refined analysis allows us to match the bound of [10] without assuming Assumption 1.

Further details are provided in Appendix E.

5.3 Attacking parameters of an SIDH-based signature scheme

In [7], Basso et al. explore instantiating SIDH-based signatures such that the endomorphism rings can be computed classically in time less than 2^λ , where λ is the security parameter. For $\lambda = 128$, they pick $p \approx 2^{218}$. For recovering the secret key, it is sufficient to recover an ephemeral secret isogeny of smooth degree $d = \ell_2^{e_2} \approx p$. Basso et al. argued that even with a smaller p , this d -Isogeny problem should be hard enough to ensure 128-bits of security. They furthermore proposed that this could be true, even when all the endomorphism rings are public.

Using the hybrid approach with trivariate Coppersmith, Benčina et al. [10] showed that when the endomorphism rings are public, the d -Isogeny problem can be solved in time roughly $d/p^{5/8} \approx p^{3/8} \approx 2^{82}$. Their analysis therefore implies that, with public endomorphism rings, the scheme offers only about 82 bits of security. When the endomorphism rings are secret, the cost of their key-recovery attack is dominated by the computation of these rings, which takes roughly $\sqrt{p} \approx 2^{109}$ time. This represents a reduction of 19 bits compared to the claimed security level for the chosen parameters. To achieve 128-bits of security, their analysis suggests that the parameters must be adjusted so that every secret isogeny has degree at least $p^{5/8}2^\lambda \approx 2^{264}$.

With Theorem 6, we improve on the state-of-the art for solving d -Isogeny for smooth d . When the endomorphism rings are public, we can recover the key in time roughly $d/p^{2/3} \approx p^{1/3} \approx 2^{73}$, reducing the security by 9-bits. To achieve 128-bit security, our analysis implies that the degrees must be increased to at least $p^{2/3}2^\lambda \approx 2^{273}$.

6 Experiments

In this section, we implement our algorithm under different parameter settings. These experiments are conducted using an Intel Xeon Platinum 8480+ CPU with SageMath 10.3. We use some experiment parameters from [10], and the details of the polynomials used can be found in Appendix F.

Our algorithm applies only to d satisfying $p^{1/2} \leq d \leq p^{2/3}$. We first run the LLL algorithm to find short vectors, and then use a Gröbner basis computation to recover the target root. For Gröbner basis, we use the CRT-based implementation from [40], which computes Gröbner bases modulo small primes and then reconstructs the result using the Chinese remainder theorem.

We summarize our results in Table 1, where we report the time for LLL and the total running time.

$\log_2 p$	$\log_2 d$	Lattice Dim.	Time for LLL (s)	Total Time (s)
253	140	70	0.71	1.68
253	150	210	16.13	21.24
253	160	70	0.77	1.03
253	165	210	19.07	24.06
500	330	210	22.72	28.84

Table 1: Experimental results for different parameters.

Discussion on Assumption 1. As discussed earlier, for multivariate Coppersmith’s method ($k > 3$), Assumption 1 is required to ensure that we can recover the target root [10, 22, 37–39, 46]. In practice, however, achieving the theoretical bounds is often difficult, since very large lattice dimensions are needed. Thus, Coppersmith’s method tends to work well for smaller parameters but its performance degrades as the parameters approach the theoretical bound.

In our experiments, we observed an interesting phenomenon: the shortest $k - 1$ vectors from the LLL-reduced basis do not always satisfy Assumption 1, and longer vectors sometimes need to be considered.

For example, when $m = 2$ in Exp. 1, we obtained 49 vectors satisfying Lemma 3. If we use only the first 3 vectors together with the original f , the ideal generated is not zero-dimensional. However, if we instead use the original f together with vectors no. 1, no. 2, and no. 49 from the LLL basis, the resulting ideal is zero-dimensional, from which we can recover the target root. One possible strategy is to run Gröbner basis computations on all vectors that satisfy Lemma 3.

The situation worsens as d gets closer to the theoretical bound. For larger d (e.g., Exp. 3), the ideal remains one-dimensional even if we use all vectors satisfying Lemma 3. However, if we include three additional longer vectors, the

ideal becomes zero-dimensional, and the root can be recovered. This suggests that longer vectors may still contain useful information.

Our experimental strategy is as follows. We begin by setting $m = 2$ and constructing the ideal from all vectors satisfying Lemma 3, and then compute its dimension. If the resulting ideal has dimension 1, we expand the set to include all available vectors (including those not satisfying Lemma 3) and test again. At that point, we perform a binary search over the ordered list of vectors to determine the smallest index j_0 (respectively, the largest index j_1) such that the ideal generated by f together with the first j_0 (resp. j_1) vectors in the LLL-reduced basis becomes zero-dimensional. We then enumerate over all ideals generated by f and the first j vectors, for every j with $j_0 \leq j \leq j_1$.

If no such pair (j_0, j_1) exists, we increment m , up to a maximum of $m = 4$. If the method still fails, we output *failed* due to the limitations of our experimental environment. If, on the other hand, the dimension of the initial ideal is strictly larger than 1 (e.g., equal to 2), we repeat the search procedure while adding additional vectors.

7 Conclusion

In this paper, we proposed an explicit formula for computing asymptotic bounds of Coppersmith’s method for integer polynomial equations. We then applied this formula to quadratic forms with Gram matrices. As an application, we improved the state-of-the-art algorithm for finding fixed-degree isogenies, and strengthened the attack of Benčina et al. on a signature scheme introduced at ACNS 2024, reducing its security by 9 bits.

Open Problem. By applying the four-variable version of Coppersmith’s method, we obtain an improved algorithm over the previously known results. Nevertheless, we believe that our bounds can be further improved, as the underlying quaternion algebra has not yet been fully exploited; we leave this as an open problem.

A closely related problem is the *embedding order problem*: let p be a prime, let $\mathcal{O}$ be a maximal order inside $B_{p,\infty}$, and let t, n be integers such that there exists an element of norm n and trace t in $\mathcal{O}$. The task is to find $\alpha \in \mathcal{O}$ with $n(\alpha) = n$ and $\text{tr}(\alpha) = t$.

In [10], it was conjectured that this problem can be solved in polynomial time whenever $n < p^{0.8}$ using trivariate Coppersmith’s method. We argue that this bound can be improved to $n < p$ by applying Coppersmith’s method more directly. More recently, Eriksen et al. [21] showed that by exploiting quaternion algebra, the problem reduces to computing traces with known residues modulo p , and demonstrated that it can be solved for $n < p^{4/3}$ even via trivial enumeration.

A natural idea is therefore to combine their approach with Coppersmith’s method. The main challenges are proving irreducibility and bounding the infinity norm. We argue that the infinity norm is on the order of p^2 , which suggests that

the bound could potentially be improved all the way to $n < p^2$. We also leave this as an open problem.

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A The extended strategy for a single polynomial equation

Often it is advantageous to use extra shifts of a certain variable; this is the Jochemsz–May extended strategy [31]. We adapt the basic strategy with extra shifts of x_1 and set $E = \{(0, \dots, 0), (1, 0, \dots, 0)\} \subseteq \mathbb{Z}^k$.

Let $\varepsilon > 0$ be arbitrarily small. Fix an integer m (depending on $1/\varepsilon$) and an integer t . Define

$$S = \bigcup_{0 \leq j \leq t} \{x_1^j \cdot x_1^{i_1} \cdots x_k^{i_k} \mid (i_1, \dots, i_k) \in (m-1)A(f)\},$$

$$M = \bigcup_{0 \leq j \leq t} \{x_1^j \cdot x_1^{i_1} \cdots x_k^{i_k} \mid (i_1, \dots, i_k) \in mA(f)\}.$$

Set $R = W \prod_{j=1}^k X_j^{l_j}$, where l_j is the largest exponent of x_j among monomials in S , and define $\tilde{f} = a_0^{-1}f \bmod R$, where a_0 is the constant term of f . Define

$$g = x_1^{i_1} \cdots x_k^{i_k} \tilde{f}(x_1, \dots, x_k) \prod_{j=1}^k X_j^{l_j - i_j} \quad \text{for } x_1^{i_1} \cdots x_k^{i_k} \in S,$$

$$\tilde{g} = x_1^{i_1} \cdots x_k^{i_k} R \quad \text{for } x_1^{i_1} \cdots x_k^{i_k} \in M \setminus S.$$

We use the coefficient vectors of $g(x_1 X_1, \dots, x_k X_k)$ and $\tilde{g}(x_1 X_1, \dots, x_k X_k)$ as a basis to construct the lattice $\mathcal{L}$.

The related formulas for asymptotic bounds are provided in Theorem 8. The proof is similar to that of Theorem 1.

Theorem 8. *For any well-defined k -variate f , the asymptotic bound obtained by the Extended Strategy is, for any arbitrarily small constant $\varepsilon > 0$:*

$$\prod_{j=1}^k X_j^{\sum_{i=0}^k b_i^{(j)} m^{k-i} t^i} < W^{\sum_{i=0}^k c_i m^{k-i} t^i - \varepsilon m^k}$$

or, writing $t = \tau m$,

$$\prod_{j=1}^k X_j^{\sum_{i=0}^k b_i^{(j)} \tau^i} < W^{\sum_{i=0}^k c_i \tau^i - \varepsilon}.$$

These coefficients satisfy the following system, determined by solving the linear system for $\lambda = 0, \dots, k+1$:

$$\begin{cases} \sum_{i=0}^k \frac{b_i^{(j)}}{k+1-i} \lambda^i = \frac{\int_{N(A(f)+\lambda E)} x_j dV}{\text{ind}(A(f)+\lambda E)}, \\ \sum_{i=0}^k c_i \lambda^i = \frac{\int_{N(A(f)+\lambda E)} 1 dV}{\text{ind}(A(f)+\lambda E)}. \end{cases}$$

Remark 10. Here $t = \tau m$ must be an integer to satisfy sumset conditions. The asymptotic bounds can thus be viewed as a function of τ . When the optimal τ_0 is irrational, one may use $t = \lceil \tau_0 m \rceil$ or $t = \lfloor \tau_0 m \rfloor$.

B Solving a system of polynomial equations with Coppersmith's method

In [10], it is shown that finding a fixed-degree isogeny reduces to solving a single integer polynomial equation when the endomorphism ring is known. In certain scenarios—such as side-channel attacks—additional information may leak and can be modeled as extra polynomial constraints, motivating a generalization from a single polynomial equation to a system. Our goal is to show that bounds derived from multiple equations can outperform those obtained from any individual equation.

Let $f_1, \dots, f_n \in \mathbb{Z}[x_1, \dots, x_k]$ be irreducible. To find all integer roots $(x_1^{(0)}, \dots, x_k^{(0)})$ such that $|x_i^{(0)}| < X_i$ for $i = 1, \dots, k$, we define

$$W_i = \|f_i(x_1 X_1, \dots, x_k X_k)\|_\infty,$$

for each $i = 1, \dots, n$. Then we construct the monomial sets

$$\begin{aligned} S &= \bigcup_{0 \leq j \leq t} \left\{ x_1^j \cdot x_1^{i_1} \cdots x_k^{i_k} \mid (i_1, \dots, i_k) \in (m-1) \bigcup_{\ell=1}^n A(f_\ell) \right\}, \\ M &= \bigcup_{0 \leq j \leq t} \left\{ x_1^j \cdot x_1^{i_1} \cdots x_k^{i_k} \mid (i_1, \dots, i_k) \in m \bigcup_{\ell=1}^n A(f_\ell) \right\}. \end{aligned}$$

Let l_j denote the largest exponent of x_j among monomials in S . For each $i = 1, \dots, n$, set

$$R_i = W_i \prod_{j=1}^k X_j^{l_j},$$

and define

$$R = \max\{R_i \mid i = 1, \dots, n\}.$$

Let $f := \sum_{i=1}^n f_i$, and define

$$\tilde{f} = f(0, \dots, 0)^{-1} f \bmod R.$$

Define the shift polynomials

$$\begin{aligned} g &= x_1^{i_1} x_2^{i_2} \cdots x_k^{i_k} \cdot \tilde{f}(x_1, \dots, x_k) \cdot \prod_{j=1}^k X_j^{l_j - i_j} \quad \text{for } x_1^{i_1} \cdots x_k^{i_k} \in S, \\ \tilde{g} &= x_1^{i_1} x_2^{i_2} \cdots x_k^{i_k} \cdot R \quad \text{for } x_1^{i_1} \cdots x_k^{i_k} \in M \setminus S. \end{aligned}$$

We derive asymptotic bounds for this system along the lines of the single-polynomial case. However, we require the existence of a constant c such that

$$R \leq c \cdot \min\{R_i \mid i = 1, \dots, n\},$$

to ensure that the $k - 1$ polynomials corresponding to short lattice vectors are independent of the choice of f_i , $i = 1, \dots, n$.

To illustrate the benefit of multiple equations, consider the example

$$\begin{aligned} f_1 &= 1 + xy + xy^2 + x^2 + x^2y^2 + z, \\ f_2 &= 1 + xy + yx^2 + y^2 + x^2y^2 + z. \end{aligned}$$

For a single equation, one obtains the bounds

$$X^{11}Y^8Z^3 < W^3 \quad \text{or} \quad X^8Y^{11}Z^3 < W^3.$$

If $X = Y = Z$, the single-equation bound becomes $X < W^{3/22}$. In contrast, our system-based strategy yields

$$X^3Y^3Z < W,$$

which for $X = Y = Z$ gives $X < W^{1/7}$, strictly improving upon the single-equation bound.

C Omitted proofs

C.1 Proof of Lemma 8

Proof. From Equation (3), the success condition for the Basic strategy is

$$2^{\frac{\dim(\mathcal{L})(\dim(\mathcal{L})-1)}{4}} \det(\mathcal{L}) \leq \left(\frac{R}{\sqrt{\dim(\mathcal{L})}} \right)^{\dim(\mathcal{L})+2-k}.$$

Equivalently,

$$\prod_{j=1}^k X_j^{p_j} < \frac{W^{p_W}}{2^{\frac{\dim(\mathcal{L})(\dim(\mathcal{L})-1)}{4}} \dim(\mathcal{L})^{\frac{\dim(\mathcal{L})+2-k}{2}}},$$

where

$$p_j = \sum_{x_1^{i_1} \cdots x_k^{i_k} \in M \setminus S} i_j, \quad p_W = |S| + 2 - k.$$

We expand p_W as

$$\begin{aligned} p_W &= \binom{(m-1)\delta + k}{k} + 2 - k \\ &= \frac{\prod_{i=0}^{k-1} ((m-1)\delta + i)}{k!} + 2 - k \\ &= \frac{\delta^k}{k!} m^k + \frac{k(k+1)\delta^{k-1} - 2k\delta^k}{2k!} m^{k-1} + O(m^{k-2}), \end{aligned}$$

and for $j = 1, \dots, k$,

$$\begin{aligned} p_j &= \binom{m\delta + k}{k+1} - \binom{(m-1)\delta + k}{k+1} \\ &= \frac{\delta^{k+1}}{k!} m^k + \frac{k^2\delta^k - k\delta^{k+1}}{2k!} m^{k-1} + O(m^{k-2}). \end{aligned}$$

The lattice dimension is

$$\dim(\mathcal{L}) = \binom{m\delta + k}{k} = \frac{\delta^k}{k!} m^k + O(m^{k-1}).$$

Thus the condition becomes

$$\left(\prod_{j=1}^k X_j \right)^{\frac{\delta^{k+1}}{k!} m^k + O(m^{k-1})} < \frac{W^{\frac{\delta^k}{k!} m^k + O(m^{k-1})}}{2^{\Theta(m^{2k})} \dim(\mathcal{L})^{\frac{\delta^k}{2k!} + o(1)}}.$$

Taking the m^k -th root, we obtain

$$\left(\prod_{j=1}^k X_j \right)^{\frac{\delta^{k+1}}{k!} + O(1/m)} < \frac{W^{\frac{\delta^k}{k!} + O(1/m)}}{2^{\Theta(m^k)} \dim(\mathcal{L})^{\frac{\delta^k}{2k!} + o(1)}}.$$

Asymptotically, this reduces to

$$\left(\prod_{j=1}^k X_j \right)^{\frac{\delta^{k+1}}{k!}} < W^{\frac{\delta^k}{k!}}.$$

The omitted terms correspond to factors of order

$$W^{\Theta(1/m)} \quad \text{and} \quad 2^{\Theta(m^k)} \dim(\mathcal{L})^{\frac{\delta^k}{2k!} + o(1)}.$$

In particular, comparing the coefficients of m^{k-1} in the expansions of p_j and p_W shows that

$$\frac{k^2\delta^k - k\delta^{k+1}}{2k!} \cdot \frac{\delta^k}{k!} > \frac{k(k+1)\delta^{k-1} - 2k\delta^k}{2k!} \cdot \frac{\delta^{k+1}}{k!},$$

which implies that there exists a constant $c_1 > 0$ such that the factor $W^{c_1/m}$ must be considered.

Moreover, the denominator contributes at least $2^{\Theta(m^k)}$, i.e., there exists $c_2 > 0$ such that we must account for a factor $2^{c_2 m^k}$. Balancing $W^{c_1/m}$ against $2^{c_2 m^k}$ (taking logarithms) yields the optimal choice $m \asymp (\log W)^{1/(k+1)}$, giving a total brute-force cost of size

$$W^{c_1/m} \cdot 2^{c_2 m^k} \geq 2^{\Omega(\log^{\frac{k}{k+1}} W)}.$$

Therefore, eliminating the ε -term in Theorem 2 necessarily requires time

$$\exp(\Theta(\log^{\frac{k}{k+1}} W)).$$

□

C.2 Proof of Lemma 9

Proof. The degree of $f_{\mathbf{G}}$ is 2. By Theorem 2, the method succeeds when

$$X_1 \cdot \dots \cdot X_k < W^{\frac{1}{2}-\varepsilon}.$$

Since the constant coefficient of $f_{\mathbf{G}}(x_1 X_1, \dots, x_k X_k)$ equals d , we have

$$W = \|f_{\mathbf{G}}(x_1 X_1, \dots, x_k X_k)\|_{\infty} \geq d.$$

Therefore, we can find all integer tuples if $X_1 \cdot \dots \cdot X_k < d^{\frac{1}{2}-\varepsilon}$.

Let g_{ij} denote the (i, j) -entry of $\mathbf{G}$. Now we exhaustively search the $\lceil \log_2 c \rceil$ most significant bits of some x_{j_0} with j_0 satisfied $X_{j_0}^2 g_{j_0 j_0} = \min_j \{X_j^2 g_{jj}\}$. For x_{j_0} , we aim to find roots with $x'_{j_0} < X_{j_0}/c$. By Lemma 1, we know the new polynomial

$$f'_{\mathbf{G}}(x_1, \dots, x'_{j_0} + M, \dots, x_k) = f_{\mathbf{G}}(x_1 X_1, \dots, x'_{j_0} \frac{X_{j_0}}{c} + M, \dots, x_k X_k),$$

is still irreducible over integers.

Suppose we already know the $\lceil \log_2 c \rceil$ most significant bits of some x_{j_0} . Now we consider

$$W' = \|f_{\mathbf{G}}(x_1 X_1, \dots, x'_{j_0} \frac{X_{j_0}}{c} + M, \dots, x_k X_k)\|_{\infty},$$

where

$$M = 2^{\lfloor \log_2 X_{j_0} \rfloor - \lceil \log_2 c \rceil} \cdot \left\lfloor \frac{X_{j_0}}{2^{\lfloor \log_2 X_{j_0} \rfloor - \lceil \log_2 c \rceil}} \right\rfloor$$

satisfies $1/2 X_{j_0} \leq M \leq X_{j_0}$.

The term $x_{j_0}^2 X_{j_0}^2 g_{j_0 j_0}$ is replaced by

$$x'_{j_0} \frac{X_{j_0}^2}{c^2} g_{j_0 j_0} + 2 x'_{j_0} \frac{X_{j_0}}{c} M g_{j_0 j_0} + M^2 g_{j_0 j_0},$$

and $x_{j_0} x_j X_{j_0} X_j g_{j_0 j}$ by

$$x'_{j_0} x_j \frac{X_{j_0}}{c} X_j g_{j_0 j} + x_j M X_j g_{j_0 j}.$$

Hence, it holds that $W' \geq d$ and $W \geq W' \geq \frac{1}{4}W$, so by Theorem 2, we can find the desired roots if

$$X_1 \cdots \frac{X_{j_0}}{c} \cdots X_k < d^{\frac{1}{2}-\varepsilon},$$

i.e.,

$$X_1 \cdots X_k < c \cdot d^{\frac{1}{2}-\varepsilon}.$$

For the time complexity, our algorithm has two steps: one is brute-forcing the $\lceil \log_2 c \rceil$ most significant bits, and the second is using Coppersmith's method. By Theorem 2, for each candidate in the first step, it costs $\text{polylog } W' = \text{polylog } W$, so the total time is $c \cdot \text{polylog } W$. $\square$

C.3 Proof of Theorem 3

Proof. We outline the strategy in four steps: first, we construct a set of upper bounds that covers every solution of interest; next, we show that every valid solution is captured by one of these bounds; then, we enumerate solutions within these bounds; and finally, we bound the running time.

1. Constructing the Bounds Set.

Consider all nonnegative integer tuples $(d_1, \dots, d_k) \in \mathbb{Z}_{\geq 0}^k$ satisfying

$$d_1 + \cdots + d_k = \left\lceil \log_2 \left(d^{\frac{1}{2}-\varepsilon} \right) + \log_2 c \right\rceil + k.$$

Denote the set of all such tuples by $\mathcal{D}$. For any $(d_1, \dots, d_k) \in \mathcal{D}$, define the corresponding upper bounds

$$(X_1, \dots, X_k) = (2^{d_1}, \dots, 2^{d_k}),$$

and let

$$\mathcal{X} = \{(2^{d_1}, \dots, 2^{d_k}) : (d_1, \dots, d_k) \in \mathcal{D}\}.$$

Thus

$$|\mathcal{X}| = \binom{\left\lceil \log_2 \left(d^{\frac{1}{2}-\varepsilon} \right) + \log_2 c \right\rceil + 2k - 1}{k - 1}.$$

Note that k is a constant and $\left\lceil \log_2 \left(d^{\frac{1}{2}-\varepsilon} \right) + \log_2 c \right\rceil + 2k - 1 < 3 \log_2 d / 2 + 2k - 1$, the size of $\mathcal{X}$ is polynomial in $\log d$.

2. Covering All Valid Solutions. Consider any solution $(x_1^{(0)}, \dots, x_k^{(0)})$ satisfying

$$\prod_{j=1}^k (|x_j^{(0)}| + 1) < c \cdot d^{\frac{1}{2}-\varepsilon}.$$

Then it holds that

$$\sum_{j=1}^k \lceil \log_2(|x_j^{(0)}| + 1) \rceil \leq \sum_{j=1}^k \log_2(|x_j^{(0)}| + 1) + k.$$

Since $\prod_{j=1}^k (|x_j^{(0)}| + 1) < c \cdot d^{\frac{1}{2}-\varepsilon}$, it follows that

$$\sum_{j=1}^k \lceil \log_2(|x_j^{(0)}| + 1) \rceil \leq \left\lceil \log_2\left(d^{\frac{1}{2}-\varepsilon}\right) + \log_2 c \right\rceil + k.$$

Hence, there exists a tuple $(X_1, \dots, X_k) \in \mathcal{X}$ such that

$$|x_j^{(0)}| < X_j \quad \text{for each } j = 1, \dots, k.$$

3. Enumeration Using Lemma 9. For any $(X_1, \dots, X_k) \in \mathcal{X}$, observe that

$$\prod_{j=1}^k X_j = 2^{d_1 + \dots + d_k} = 2^{\lceil \log_2(d^{\frac{1}{2}-\varepsilon}) + \log_2 c \rceil + k} \leq c \cdot 2^{k+1} d^{\frac{1}{2}-\varepsilon}.$$

As we want to find all solutions $(x_1^{(0)}, \dots, x_k^{(0)})$ satisfying

$$f_{\mathbf{G}}(x_1^{(0)}, \dots, x_k^{(0)}) = 0 \quad \text{and} \quad |x_j^{(0)}| \leq X_j \quad \text{for all } j,$$

where

$$\prod_{j=1}^k X_j \leq c \cdot 2^{k+1} \cdot d^{\frac{1}{2}-\varepsilon}.$$

By Lemma 9 and k is a constant, it costs in time $c \cdot \text{polylog}(\|f_{\mathbf{G}}(x_1 X_1, \dots, x_k X_k)\|_{\infty})$.

4. Bounding the Running Time.

Define

$$W_{\max} := \max_{(d_1, \dots, d_k) \in \mathcal{D}} \|f_{\mathbf{G}}(x_1 2^{d_1}, \dots, x_k 2^{d_k})\|_{\infty}.$$

Write $g_{ij} = \langle \mathbf{b}_i, \mathbf{b}_j \rangle$ for the entries of $\mathbf{G}$. Set

$$\mathcal{I}_1 = \max_{i \neq j} \{ 2^{d_i + d_j} |\langle \mathbf{b}_i, \mathbf{b}_j \rangle| \},$$

$$\mathcal{I}_2 = \max_i \{ 2^{2d_i} \|\mathbf{b}_i\|_2^2 \},$$

$$\mathcal{I}_3 = d.$$

Then,

$$\|f_{\mathbf{G}}(x_1 2^{d_1}, \dots, x_k 2^{d_k})\|_{\infty} = \max\{\mathcal{I}_1, \mathcal{I}_2, \mathcal{I}_3\}.$$

By the Cauchy–Schwarz inequality, for $i \neq j$,

$$(2^{d_i + d_j} |\langle \mathbf{b}_i, \mathbf{b}_j \rangle|)^2 \leq (2^{2d_i} \|\mathbf{b}_i\|_2^2) (2^{2d_j} \|\mathbf{b}_j\|_2^2),$$

so that $\mathcal{I}_1 \leq \mathcal{I}_2$. Next, we bound $\mathcal{I}_2$ using a crude estimate:

$$\begin{aligned}
\mathcal{I}_2 &\leq \prod_{i=1}^k (2^{2d_i} \|\mathbf{b}_i\|_2^2)^{1/k} \cdot 2^{2 \sum_{i=1}^k d_i - \frac{2}{k} \sum_{i=1}^k d_i} \\
&\leq 2^{2 \sum_{i=1}^k d_i} \cdot \prod_{i=1}^k \|\mathbf{b}_i\|_2^2 \\
&= 2^{2 \lceil \log_2 d^{\frac{1}{2}-\varepsilon} \rceil + 2k} \cdot \prod_{i=1}^k \|\mathbf{b}_i\|_2^2 \\
&\leq 2^{2k+2} d \cdot 2^{k(k-1)} \det(\mathbf{G}) \quad (\text{since } \prod_{i=1}^k \|\mathbf{b}_i\|_2^2 \leq 2^{k(k-1)} \det(\mathbf{G})) \\
&= 2^{k^2+k+2} d \det(\mathbf{G}).
\end{aligned}$$

Therefore,

$$W_{\max} \leq \max\{2^{k^2+k+2} d \det(\mathbf{G}), d\}.$$

Since $\det(\mathbf{G})$ (and hence $\log \det(\mathbf{G})$) is bounded in terms of $\|\mathbf{G}\|_\infty$, it follows that $\log W_{\max}$ is polynomial in $\log d$ and $\log \|\mathbf{G}\|_\infty$.

In summary, by considering all tuples in $\mathcal{X}$ and applying Lemma 9 on each, we recover every solution satisfying

$$\prod_{j=1}^k (|x_j^{(0)}| + 1) < c \cdot d^{\frac{1}{2}-\varepsilon}$$

in time $c \cdot \text{polylog}(d, \|\mathbf{G}\|_\infty)$. □

D Minor issues in previous work

In this section, we elaborate on a couple of minor issues in [10].

D.1 The omitted sub-exponential factor

In this section, we give a detailed explanation of why [10] requires an additional sub-exponential factor. We begin by recalling the Coppersmith-type method used in Theorem 6.2 and Proposition 7.1 of [10], which is based on a trivariate variant due to Bauer and Coron [9].

Bauer and Coron employed a trivariate variant of admissible sets (cf. Definition 5), extending the bivariate version introduced by Blömer and May [11]. This can be generalized to the multivariate case as follows:

Definition 5 (Admissible sets [9, 11]). Let $f(x_1, \dots, x_k) \in \mathbb{Z}[x_1, \dots, x_k]$ be a multivariate integer polynomial, and let S, M be finite non-empty sets of monomials in the variables $x_1, \dots, x_k$. We say that f is defined over S if $\text{supp}(f) \subseteq S$. We say that S and M are *admissible* for f if:

1. For every monomial $\alpha \in S$, the polynomial $\alpha \cdot f$ is defined over M .
2. For every polynomial g defined over M , if $g = f \cdot p$ for some polynomial p , then p is defined over S .

Using admissible sets, Bauer and Coron proved the following lemma (presented as Theorem 2.5 in [10]):

Lemma 11 (Theorem 1, [9]). *Let $f \in \mathbb{Z}[x, y, z]$ be an irreducible polynomial with an integral root (x_0, y_0, z_0) satisfying $|x_0| < X$, $|y_0| < Y$ and $|z_0| < Z$. If S and M are admissible sets for f , then one can compute in polynomial time a polynomial $f_1 \in \mathbb{Z}[x, y, z]$ that has (x_0, y_0, z_0) as a root and is algebraically independent from f , provided that*

$$X^{s_x} Y^{s_y} Z^{s_z} < W^{|S|} 2^{-(6+c)|S|(d_x^2 + d_y^2 + d_z^2)}$$

and

$$(|M| - |S|)^2 \leq c|S|(d_x^2 + d_y^2 + d_z^2)$$

for some constant c . Here $W := \|f(xX, yY, zZ)\|_\infty$, and d_x, d_y, d_z denote the maximal degrees of f in x, y, z , respectively. Moreover, $s_x := \sum_{x^i y^j z^k \in M \setminus S} i$, with analogous definitions for s_y and s_z .

In [10], Lemma 11 is applied to trivariate polynomials of total degree 2, with

$$S = \{x^i y^j z^k : i + j + k \leq m\}.$$

In their asymptotic analysis, the factor $2^{-(6+c)|S|(d_x^2 + d_y^2 + d_z^2)}$ was omitted, leaving only

$$X^{s_x} Y^{s_y} Z^{s_z} < W^{|S|},$$

and then taking the limit $m \rightarrow \infty$.

However, following the derivation in Appendix A.3 of [9], the condition comes from

$$\sqrt{|M| - |S|} < 2^{-\frac{|M| - |S| - 1}{4}} \left(2^{-6|S|(d_x^2 + d_y^2 + d_z^2)} W^{|S|} X^{-s_x} Y^{-s_y} Z^{-s_z} \right)^{\frac{1}{|M| - |S|}},$$

which can be rearranged as

$$X^{s_x} Y^{s_y} Z^{s_z} < \frac{W^{|S|}}{2^{\frac{(|M| - |S|)(|M| - |S| - 1) + 24|S|(d_x^2 + d_y^2 + d_z^2)}{4}} (|M| - |S|)^{\frac{|M| - |S|}{2}}}. \quad (6)$$

For Theorem 6.2 and Proposition 7.1 in [10], the relevant polynomial has total degree 2, hence its support consists of the monomials $\{x^i y^j z^k : i + j + k \leq 2\}$. With $S = \{x^i y^j z^k : i + j + k \leq m\}$, Property 1 of Definition 5 ensures that

$$\{x^i y^j z^k : i + j + k \leq m + 2\} \subseteq M,$$

implying $|M| - |S| = \Omega(m^3)$.

Moreover, from the last line of Table 1 in [10], we see that

$$s_x = s_y = s_z = \sum_{i=0}^{m+1} (m+2-i)(i+1) + \sum_{i=0}^m (m+1-i)(i+1),$$

from which it follows that $M = \{x^i y^j z^k : i+j+k \leq m+2\}$.

We then compute:

$$|S| = \frac{1}{6}m^3 + m^2 + O(m), \quad s_x = s_y = s_z = \frac{1}{3}m^3 + \frac{5}{2}m^2 + O(m).$$

Plugging into (6) and normalizing by taking the m^3 -rd root, we obtain

$$(XYZ)^{\frac{1}{6}+O(1/m)} < \frac{W^{\frac{1}{3}+O(1/m)}}{2^{\Theta(m^3)} \cdot 2^{\Theta(\log m)}}.$$

Asymptotically, this reduces to

$$XYZ < W^{1/2},$$

but with omitted terms of order $W^{\Theta(1/m)}$ and $2^{\Theta(m^3)}$. If we take $m = \lfloor 1/\varepsilon \rfloor$, then it becomes

$$XYZ < W^{1/2-\varepsilon}.$$

Now we show eliminating the ε -term necessarily incurs an extra cost of $\exp(O(\log^{3/4} W))$. Similar to Lemma 8, whenever we obtain a bound $W^{1/2}$ on the size of our solution, we can enlarge this bound to $cW^{1/2}$ by performing a brute-force search. This search increases the time complexity by a factor of c and we will show $c = 2^{\Omega(\log^{3/4} W)}$.

Comparing the coefficients of m^2 in $|S|$ and s_x shows that

$$\frac{5}{2} \cdot \frac{1}{6} > 1 \cdot \frac{1}{3},$$

which implies that a factor $W^{c_1/m}$ must be accounted for, for some constant $c_1 > 0$. Furthermore, the denominator contributes at least $2^{c_2 m^3}$ for some $c_2 > 0$.

Balancing $W^{c_1/m}$ with $2^{c_2 m^3}$ (after taking logarithms) gives the optimal choice $m \asymp (\log W)^{1/4}$, which yields a brute-force cost of size

$$W^{c_1/m} \cdot 2^{c_2 m^3} \geq 2^{\Omega(\log^{3/4} W)}.$$

Conclusion. Thus, eliminating the ε -term in Theorem 6.2 and Proposition 7.1 of [10] necessarily incurs an extra cost of

$$\exp(O(\log^{3/4} W)),$$

which is precisely the claimed sub-exponential overhead.

D.2 The reducibility of the polynomial after guessing

The strategy of Benčina *et al.* of guessing one or two variables and then applying Coppersmith—can fail in a non-trivial way, because the resulting polynomial may be divisible by a large integer. Dividing by the gcd shrinks the parameter W and may make the Coppersmith condition impossible to satisfy. We give two concrete examples of this where this can happen, even when their assumption is satisfied (Assumption 2). Although we expect that this will rarely happen in practice, it is nonetheless a corner case that should be highlighted.

Example 1. Let $M \approx p^{0.1}$ and define

$$d = M^9 + M^8(M + 1) + 2M^4(M^5 + 1) = 4M^9 + M^8 + 2M^4 \approx p^{0.9}.$$

Set the variable upper bounds

$$X_1 = X_2 = X_3 = X_4 = p^{0.2}.$$

Assume the (LLL-reduced) Gram matrix $\mathbf{G}$ of the connection ideal is diagonal with entries

$$g_{11} = M^5, \quad g_{22} = M^4(M + 1), \quad g_{33} = M^5 + 1, \quad g_{44} = M^5 + 1,$$

so that $g_{11} \approx g_{22} \approx g_{33} \approx g_{44} \approx p^{0.5}$, in accordance with Assumption 2. Note furthermore that $\gcd(g_{11}, g_{22}, g_{33}, g_{44}) = 1$.

Observe that (M^2, M^2, M^2, M^2) is a solution since

$$\begin{aligned} \sum_{i=1}^4 g_{ii} x_i^2 &= g_{11} M^4 + g_{22} M^4 + g_{33} M^4 + g_{44} M^4 \\ &= M^9 + (M^9 + M^8) + (M^9 + M^4) + (M^9 + M^4) \\ &= 4M^9 + M^8 + 2M^4 \\ &= d. \end{aligned}$$

Now suppose an attacker guesses x_3 and x_4 (here the correct values M^2), and thereby reduces to a bivariate equation in x_1, x_2 . The equation becomes

$$g_{11} x_1^2 + g_{22} x_2^2 = d - g_{33} x_3^2 - g_{44} x_4^2.$$

Plugging $x_3 = x_4 = M^2$ and simplifying yields

$$M^5 x_1^2 + M^4(M + 1) x_2^2 = 2M^9 + M^8,$$

or equivalently the bivariate polynomial

$$g(x_1, x_2) = M^5 x_1^2 + M^4(M + 1) x_2^2 - 2M^9 - M^8.$$

The coefficients of g have a nontrivial common factor M^4 . Dividing by that gcd gives

$$g'(x_1, x_2) = M x_1^2 + (M + 1) x_2^2 - 2M^5 - M^4.$$

We now compare the Coppersmith parameter W before and after dividing by the gcd. With $X_1 = X_2 = p^{0.2}$ and $M \approx p^{0.1}$,

$$\|g(x_1 X_1, x_2 X_2)\|_\infty \approx p^{0.9},$$

whereas after dividing by M^4 ,

$$\|g'(x_1 X_1, x_2 X_2)\|_\infty \approx M \cdot X_1^2 \approx p^{0.1} \cdot p^{0.4} = p^{0.5}.$$

Bivariate Coppersmith (Lemma 12) requires that

$$X_1 X_2 < W^{1/\delta} \quad (\text{here } \delta = 2),$$

i.e. $X_1 X_2 < \sqrt{W}$. After dividing by M^4 we get

$$X_1 X_2 \approx p^{0.4} \quad \text{but} \quad \sqrt{W'} \approx p^{0.25},$$

so the required inequality $p^{0.4} < p^{0.25}$ fails badly. Consequently the bivariate Coppersmith method cannot find the target roots in this scenario.

This simple example demonstrates a real obstruction: guessing some variables and then applying bivariate Coppersmith can lead to a polynomial whose coefficients have a large common factor. After dividing by that factor the effective W is drastically reduced and the Coppersmith inequality can no longer be satisfied. Hence this strategy is not sound in general without careful handling (e.g. ensuring no large gcd appears, or analyzing how guessing affects W and δ).

Example 2. Let $M \approx p^{1/120}$ and define

$$d = 4M^{78} + 4M^{18}.$$

Set

$$X_1 = X_2 = X_3 = X_4 = p^{3/40} < p^{1/12}.$$

Assume the (LLL-reduced) Gram matrix $\mathbf{G}$ of the connection ideal is diagonal with entries

$$g_{11} = M^{60} + 1, \quad g_{22} = M^{60} + 1, \quad g_{33} = M^{60}, \quad g_{44} = M^{60} + 2,$$

so that $g_{jj} \approx p^{1/2}$, in accordance with Assumption 2, and $\gcd(g_{11}, g_{22}, g_{33}, g_{44}) = 1$. Observe that (M^9, M^9, M^9, M^9) is a correct solution.

Suppose an attacker guesses x_3 and x_4 (here the correct value M^9), and thereby reduces to a bivariate equation in (x_1, x_2) . Plugging in $x_3 = x_4 = M^9$ and simplifying yields a bivariate polynomial $g(x_1, x_2)$ whose coefficients have a nontrivial common factor $M^{60} + 1$. Dividing by this gcd we obtain

$$g'(x_1, x_2) = x_1^2 + x_2^2 - 2M^{18}.$$

We now compare the Coppersmith parameter W before and after dividing out the gcd. With $X_1 = X_2 = p^{3/40}$ and $M \approx p^{1/120}$,

$$\|g(x_1 X_1, x_2 X_2)\|_\infty \approx p^{13/20},$$

whereas

$$\|g'(x_1X_1, x_2X_2)\|_\infty \approx p^{3/20}.$$

The bivariate Coppersmith condition (Lemma 12) requires

$$X_1X_2 < W^{1/2}.$$

After dividing by the gcd we have

$$X_1X_2 \approx p^{3/20} \quad \text{but} \quad \sqrt{W'} \approx p^{3/40},$$

so the required inequality $p^{3/20} < p^{3/40}$ fails badly.

However, as $d \approx p^{13/20} < p^{2/3}$, the original four-variate instance *is* solvable by the four-variate Coppersmith method. Thus the two-variable guessing strategy misses a valid target solution due to reducibility of the resulting bivariate polynomial.

E Solving d -Isogeny with Bivariate Coppersmith's method

In this section, we present results obtained using the bivariate Coppersmith method of [15], which does *not* require Assumption 1 and does not incur any sub-exponential terms. We obtain two results:

1. We solve d -Isogeny for general large d . Our bound improves upon exhaustive search in the regime $p < d < p^2$, though it remains weaker than the Cornacchia-based method of [10].
2. We solve d -Isogeny with smooth d in the quantum setting. With a refined analysis, we match the bound of [10] *without* invoking Assumption 1.

We rely on the following theorem for bivariate polynomials of total degree 2 [15].

Lemma 12 (Theorem 3 of [15]). *Let $f(x, y) \in \mathbb{Z}[x, y]$ be an irreducible polynomial whose total degree is at most 2. Let $X, Y > 0$ be upper bounds on the desired integer solution (x_0, y_0) , and define*

$$W = \|f(xX, yY)\|_\infty.$$

If

$$XY < W^{1/2},$$

then one can find all integer pairs (x_0, y_0) satisfying $f(x_0, y_0) = 0$ and $|x_0| \leq X$, $|y_0| \leq Y$ in time $\text{polylog}(W)$.

Although the original polynomial in four variables is irreducible, one must check that after guessing two variables, the resulting bivariate polynomial remains irreducible. While this holds generically, it can fail in special cases (see Appendix D.2). Hence, throughout this section we impose the following assumption.

Assumption 3. *After guessing one or two variables, the resulting polynomial remains irreducible over the integers.*

We now apply Lemma 12 to solve d -Isogeny for general large d .

Theorem 9. *Let $d = p^{1/2+\delta}$ be an integer, where $\delta > 0$ is a constant. Let E_1, E_2 be d -isogenous supersingular elliptic curves over $\mathbb{F}_{p^2}$. Under Assumption 2 and Assumption 3, there exists a classical algorithm for d -Isogeny running in time*

$$\tilde{O}(\max\{p^{1/2}, p^{3\delta/2-1/4}\}).$$

Proof. We follow the usual blueprint and apply Coppersmith's method to find a root of $f_{\mathbf{G}}$. We guess two of the four variables, along with the most significant bits of the remaining two variables, so that the total unknown least significant bits have at most $\log_2 d/2$ bits.

Since each variable has size $\tilde{O}(p^{\delta/2})$, the guessing cost is

$$\tilde{O}(p^{2\delta}/\sqrt{d}) = \tilde{O}(p^{3\delta/2-1/4}).$$

By Assumption 3 and Lemma 1, the resulting polynomial is still irreducible. We then solve the two unknown least significant bits in polynomial time by using Lemma 12. Computing the endomorphism ring costs $\tilde{O}(p^{1/2})$ time. Thus the overall running time is

$$\tilde{O}(\max\{p^{1/2}, p^{3\delta/2-1/4}\}).$$

□

Remark 11. *This improves over exhaustive search whenever $p < d < p^2$, but remains weaker than the Cornacchia-based method of [10]. Our bound requires no assumptions beyond Assumption 2. Note also that we use the original bivariate Coppersmith method of [15], and not the version of [16] used in [10]. Therefore we avoid the sub-exponential factor appearing in their analysis.*

Theorem 10. *Let $d = p^{1/2+\delta}$ be a smooth integer, where $\delta > 0$ is a constant. Let E_1, E_2 be d -isogenous supersingular elliptic curves over $\mathbb{F}_{p^2}$. Under Assumption 2 and Assumption 3, there exists a quantum algorithm for d -Isogeny running in time*

$$\tilde{O}(\max\{p^{1/4}, p^{\delta/2}\}).$$

Proof. Using Grover search provides a quadratic speedup for the guessing step. As in Theorem 9, we decompose the degree as $d = d_{\text{guess}} \cdot d_{\text{solve}}$, where $d_{\text{guess}} = \tilde{O}(p^{\delta-1/2})$. We enumerate possible isogenies φ_{guess} of degree d_{guess} . The remaining component φ_{solve} has degree $d_{\text{solve}} = \tilde{O}(p)$ and can be recovered in polynomial time using Lemma 12 after guessing two variables under Assumption 2 and Assumption 3.

The guessing space has size $\tilde{O}(p^\delta)$, which becomes $\tilde{O}(p^{\delta/2})$ under Grover search. Computing the endomorphism ring costs $\tilde{O}(p^{1/4})$. Thus the total running time is

$$\tilde{O}(\max\{p^{1/4}, p^{\delta/2}\}).$$

□

Remark 12. *The quantum algorithm of [10] (Section 7), which relies on trivariate Coppersmith under Assumption 1, can also be improved by using this refined analysis. However, its complexity remains worse than that of Theorem 10.*

F Polynomials used in our experiments

We provide the explicit polynomials used in our experiments. All polynomials are quadratic forms in (x_1, x_2, x_3, x_4) with large integer coefficients.

Exp. 1

$$\begin{aligned} f = & 40473403944771042310329985208052973550 x_1^2 + 9748251956265421211829331742284787136 x_1 x_2 \\ & + 77454633550189127124614625500468942822 x_2^2 + 9748251956265421211829331742284787136 x_1 x_3 \\ & + 48985934264245616245264531878868621177 x_2 x_3 + 902419591443790462258239566163618070005 x_3^2 \\ & + 24976524946590553383304748705963317367 x_1 x_4 - 9748251956265421211829331742284787136 x_2 x_4 \\ & + 214461543037839266660245298330265316992 x_3 x_4 + 1716006642762988328172501652596769393989 x_4^2 \\ & - 1393796574908163946345982392040522594123776. \end{aligned}$$

Exp. 2

$$\begin{aligned} f = & 3897547560480989392878390612806074489 x_1^2 + 89643593891062756036202984094539713247 x_2^2 \\ & + 2549863555395826816290179291063255984 x_1 x_3 + 49087054174737412952161615483258835949 x_2 x_3 \\ & + 774065686297346099914796435657452606484 x_3^2 - 2316483448965540237620928129585942081 x_1 x_4 \\ & + 58646861774104016774674123694454887632 x_2 x_4 + 15299181332374960897741075746379535904 x_3 x_4 \\ & + 17649300171967851438470970389282875615042 x_4^2 - 1427247692705959881058285969449495136382746624. \end{aligned}$$

Exp. 3

$$\begin{aligned} f = & 120864756928231872170197654963601902031 x_1^2 + 40379784023742363682691476922140013988 x_1 x_2 \\ & + 194680861828862664977080049082731556942 x_2^2 - 80210377761494289609629402238643748110 x_1 x_3 \\ & + 156088821196201368396536230067211356461 x_2 x_3 + 410215661800567766037607854071586163516 x_3^2 \\ & - 49019009704082514622267696968581725297 x_1 x_4 + 147083019515271147858011236632623029956 x_2 x_4 \\ & + 205107830900283883018803927035793081758 x_3 x_4 + 615323492700851649056411781107379245274 x_4^2 \\ & - 1461501637330902918203684832716283019655932542976. \end{aligned}$$

Exp. 4

$$\begin{aligned}
f = & 138247101726295456755875073486750847154 x_1^2 + 21460434255412846014100199109945618216 x_1 x_2 \\
& + 212291917471350331602636068408707319455 x_2^2 - 7153478085137615338033399703315206072 x_1 x_3 \\
& + 12846160333952246577905930722776062645 x_2 x_3 + 325428104581287603783196981068692664128 x_3^2 \\
& + 29955363653541474694836788320420187257 x_1 x_4 + 14306956170275230676066799406630412144 x_2 x_4 \\
& + 50074346595963307366233797923206442504 x_3 x_4 + 501050875206347183111794445486198453599 x_4^2 \\
& - 46768052394588893382517914646921056628989841375232.
\end{aligned}$$

Exp. 5

$$\begin{aligned}
f = & 1476039848139568345957398233132065551070118393558818672086906013422308893138 x_1^2 \\
& + 467183475178706746354846443735702396922025500428240632379497048864323244128 x_1 x_2 \\
& + 1788278465337625016639908319382448772211377472962964127041821744241368380148 x_2^2 \\
& + 1476039848139568345957398233132065551070118393558818672086906013422308893138 x_1 x_3 \\
& + 470371281524582696419211290048484569718965391011442671123108113243249718139 x_2 x_3 \\
& + 4428119544418705037872194699396196653210355180676456016260718040266926679414 x_3^2 \\
& + 1005668566614985649538186943083580981351153002547376000963797900179059174999 x_1 x_4 \\
& - 1554686727748271643462485097514597573750364722748843810852073219809206758084 x_2 x_4 \\
& + 1669199068081321715624027157724655396582170307052688562058836040060874460814 x_3 x_4 \\
& + 5498659717285475788199468871406120419542179321772875713585637679458057504659 x_4^2 \\
& - 21872507247830119243725022271176213653531694308932124364257706064099529991993759 \\
& \quad 23223513177023053824.
\end{aligned}$$